

The Value of Clean Water: Experimental Evidence from Rural India[†]

By FIONA BURLIG, AMIR JINA, AND ANANT SUDARSHAN*

Over 2 billion people lack clean drinking water. Existing solutions face high costs (piped water) or low demand (point-of-use chlorine). Using a 60,000 household cluster-randomized experiment, we test an alternative approach: decentralized treatment and home delivery of clean water to the rural poor. At low prices, take-up exceeds 90 percent, sustained throughout the experiment. High prices reduce take-up but are privately profitable. We experimentally recover revealed-preference measures of valuation. Willingness-to-pay is several times higher than prior indirect estimates; willingness-to-accept is larger and exceeds marginal cost. Self-reported health measures improve accordingly. On a cost-per-DALY basis, free water delivery regimes appear highly cost effective. (JEL I12, O12, O13, O18, Q25, Q51, Q53)

More than 2 billion people still do not have safe drinking water at home. The costs of this deprivation are staggering. Drinking contaminated water causes approximately 2 billion cases of diarrhea and half a million deaths among children under five annually, making it the fifth-leading driver of child mortality worldwide (WHO, UNICEF, World Bank 2022).¹ Nor is this public health crisis close to being solved, with the United Nations estimating that achieving universal access to clean water by 2030 will require the pace of drinking water improvement to accelerate sixfold, even as climate

*Burlig: Harris School of Public Policy and Energy Policy Institute (EPIC) (email: burlig@uchicago.edu); Jina: Harris School of Public Policy and EPIC, University of Chicago, and NBER (email: amirjina@uchicago.edu); Sudarshan: Department of Economics, University of Warwick (email: anant.sudarshan@warwick.ac.uk). Rema Hanna was the coeditor for this article. We thank Kishan Nanavati and Krupa Paltasingh of Spring Health Water India Pvt. Ltd. for their partnership on this project. We are grateful to several anonymous referees for their feedback. We thank Kashif Ahmed, Sanghwa Ahn, Zoya Khan, Alina Gafanova, Ambalika Khanna, Nisha Koppa, Sreya Majumdar, Yashaswi Mohanty, Suyash Nandgaonkar, Chinmaya Sahoo, and Camellia Ye for excellent research assistance. We received helpful comments from Susanna Berkouwer, Chris Costello, Josh Dean, Diva Dhar, Pascaline Dupas, James Fenske, Rachel Glennerster, Michael Greenstone, Sylvan Herskowitz, Reshmaan Hussam, Koichiro Ito, Kelsey Jack, Namrata Kala, Supreet Kaur, Erin Kelley, Michael Kremer, Ashley Langer, Nadia Lucas, Charles Noussair, Chris Udry, Catherine Wolfram, and numerous seminar participants. We gratefully acknowledge financial support from the Oak Foundation, the Templeton World Charity Foundation, and the Becker Friedman Institute's Development Economics Research Fund. This project received IRB approval from the University of Chicago (protocol IRB22-0036 and IRB23-1363), and is registered on the AEA RCT registry (Burlig, Jina, and Sudarshan 2023). Replication data are available at ICPSR (Burlig, Jina, and Sudarshan 2026). All remaining errors are our own.

[†]Go to <https://doi.org/10.1257/aer.20250278> to visit the article page for additional materials and author disclosure statement(s).

¹For a sense of scale, the *annual* losses in disability-adjusted life years (DALYs) from unclean drinking water well exceed those from the first year of the global COVID pandemic.

change is exacerbating water scarcity, leaving billions of people newly vulnerable to waterborne disease (United Nations 2022; World Bank 2016).²

The aspirational “gold standard” is universal access to clean water at home, delivered through pipes and taps. Piped drinking water is not a new technology—the United Kingdom has required that new houses have clean piped water since the passage of the Public Health Act of 1875—and the dramatic expansion of purified water delivered to homes through pipes in developed countries was one of the greatest public health achievements of the nineteenth and twentieth centuries (Alsan and Goldin 2019).³ However, nearly 150 years on, less than 30 percent of the overall population in low-income countries—and only 14 percent of the rural population—has access to clean water at home (WHO, UNICEF 2024). Moreover, lab testing reveals that piped water in low-income countries is often just as contaminated as untreated surface water (World Bank 2018). In 2021, the US Centers for Disease Control and Prevention identified only 58 countries worldwide where tap water is safe for drinking. This list excludes most of Asia (including India and China), much of Central and South America, and all of sub-Saharan Africa (Centers for Disease Control and Prevention 2021). Even in India’s National Capital Region, only 60 percent of piped water samples were deemed fit to drink (Jalan and Somanathan 2008).

These facts establish two goals for economists. The first is to evaluate improved solutions that bring us closer to universal safe drinking water. The second is to quantify how much the poor value clean water in order to inform the optimal allocation of limited state funds. This paper contributes on both fronts. We use a cluster-randomized field trial, covering about 60,000 households in 120 villages in rural Odisha in India, to evaluate the demand for drinking water provided using a new approach—decentralized water treatment combined with home delivery in sealed, reusable containers (Burlig, Jina, and Sudarshan 2023). We show that this alternative could significantly improve access and we use the experiment to recover direct measures of household valuation for clean water.

Our experiment adds to a growing body of work in the economics literature that has both studied options to improve the take-up and efficacy of piped water (Galiani, Gertler, and Schargrotsky 2005; Gamper-Rabindran, Khan, and Timmins 2010; Devoto et al. 2012; Szabó 2015; dos Santos and Guidetti 2024), and evaluated cheaper alternatives suitable for low-income countries. The approach that has arguably gained the most traction has been point-of-collection or point-of-use treatment using chlorine tablets or solution. A rich body of evidence suggests that chlorine-based water treatment is both effective at improving health and extremely cheap (Kremer et al. 2011b; Luoto et al. 2011; Dupas et al. 2016; Null et al. 2018; Haushofer et al. 2021; Dupas et al. 2023). These studies and others are summarized in a review (Clasen et al. 2015) and meta-analysis (Kremer et al. 2023).⁴

² The costs of these necessary investments are substantial: Achieving global universal clean water and sanitation access by 2030 is expected to require increasing annual spending by between \$131 and \$141 billion (World Bank 2024).

³ As of 2022, 94.3 percent of the population in high-income countries had “safely managed” drinking water, defined as having an improved water source located on the premises with water that is free from contamination, available when needed (WHO, UNICEF 2024).

⁴ A smaller number of papers measure the effectiveness of alternative home treatment methods, such as water filtration (Berry, Fischer, and Guiteras 2020) or solar disinfection (Conroy et al. 1996).

Yet despite its benefits, point-of-use and point-of-collection treatment have proved insufficient to fully solve the clean water access problem in low-income countries. The Kremer et al. (2023) meta-analysis reports average point-of-use take-up rates of just 46 percent. Crucially, low demand for chlorine tablets or solutions is not due to monetary costs: Even at *zero price*, their usage has been well below universal access.⁵ Moreover, many households that do obtain chlorine do not ultimately use it to treat their drinking water.⁶ As a result, although both piped water and point-of-use chlorine have increased clean water access somewhat, overall progress is far slower than we might wish, lagging population growth in many parts of the world. Indeed, the WHO estimates that the number of people without clean water in low-income countries *increased* by 197 million between 2000 and 2022 (WHO, UNICEF 2024).

The intervention we study (home delivered treated water) provides a third option. Private sector variants of this approach are gaining increasing traction, serving a growing segment of households in urban slums and rural areas that are willing to pay for potable water (Cohen and Ray 2018; Advani et al. 2011; Brown et al. 2011; Daly et al. 2021). Between 2005 and 2015, usage of privately supplied clean water in Brazil, China, India, Indonesia, Mexico, and Thailand increased by 175 percent (Cohen and Ray 2018). Yet despite evident latent interest, to our knowledge, there exists no rigorous evidence quantifying the demand for low-cost interventions that directly supply safe water—as distinct from point-of-use treatment options—to the rural poor.

Our study location, Odisha, is one of the poorest states in India and has low levels of safe drinking water access. We conducted our study in partnership with a private company operating in rural Odisha. Section I provides more detail on our study setting and on how water was treated and delivered to the home. In the experiment, which we describe in detail in Section II, we randomize villages into one of three treatment arms: (i) a “prices” regime where water was sold at varying prices; (ii) a “free ration” regime where households could order up to 400 liters (L) of water per month for free; and (iii) an “exchangeable entitlement” regime where households could either order treated water for free or, by forgoing an order, receive cash rebates of varying amounts for every unused unit of their monthly entitlement.

Within each village, we randomized households into control or treatment. Control households could buy water at the prevailing high price charged by the company prior to the experiment. In the price arm, treated households were randomly assigned to receive discounts of 10 percent, 50 percent, or 90 percent of this price. In the exchangeable entitlement arm, treated households were correspondingly randomized to receive cash rebates equal to 10 percent, 50 percent, 90 percent, or 100 percent of this price for every bottle of their 400 L entitlement that they did

⁵Nonmonetary costs include unpleasant taste (Jeuland et al. 2016; Crider et al. 2018; Puget et al. 2010; Smith et al. 2021; Dupas et al. 2016) and the inconvenience and cognitive burden of treating water at home. Point-of-collection treatment has been found to also face contamination risks associated with households collecting and transporting their own water (Kremer et al. 2011b).

⁶These results are very consistent across contexts. Low initial take-up at low or zero prices has been documented in Bangladesh (Luoto et al. 2011), Malawi (Dupas et al. 2023), and Kenya (Null et al. 2018), while meaningful gaps between initial take-up and follow-up chlorine usage has been shown in Bangladesh (Luoto et al. 2011), Kenya (Dupas et al. 2016), and Zambia (Ashraf, Berry, and Shapiro 2010). In our own data, only 3.6 percent of representative control-group households report using chlorine to treat water in an average survey round, even though chlorine is widely available and very cheap.

not order. We analyze this experiment using rich administrative data on water orders and several household surveys, which we describe in Section III.

Our first main finding, documented in Section IV, is that home delivery of treated water can work well to expand clean water access, but households are quite price sensitive. We find that take-up of clean water at low prices is nearly universal and is sustained over the experiment duration. Over 90 percent of households order water when it is free. Take-up is similarly high at low prices: At our lowest price of 0.14 Indian rupees (Rs) per liter, take-up is about 89 percent. Strikingly, the demand for home-delivered water is high, even though households have access to cheap chlorine solution in local stores and about 34 percent of our sample report having access to piped water at some point during the experiment. Home delivery of clean water, thus, overcomes take-up ceilings documented in other water-treatment approaches (Dupas et al. 2023).

However, total demand falls substantially as prices rise. Interestingly, this variation stems largely from differences in take-up and not from differences in the amounts ordered conditional on take-up. Across all treatment arms and all prices, we find that *if* a household chooses to order water at all, they do so in substantial quantities, sufficient to fulfill all or most of their drinking water needs, based on conventional benchmarks of 1.5–2 L per day per person. For households choosing to order, demand is also stable over time, unlike prior evidence on chlorine-based water treatment (Ashraf, Berry, and Shapiro 2010; Berry, Fischer, and Guiteras 2020; Dupas et al. 2016). In addition, demand is not unbounded even when water is free. Our ration was generous enough to exceed the likely drinking water needs of most households. Although households order more water when it is free than under status-quo pricing, the ration does not bind. This suggests that a policy to provide free drinking water might generate limited waste, at least for rural populations similar to our experiment, which lack safe *drinking* water but have plentiful access to water for other purposes.

The patterns of demand we observe are consistent with informed consumers seeking to replace most of their drinking water needs with home-delivered clean water, with consequent time savings and health benefits. The latter are well documented in the literature (e.g., Kremer et al. 2023), and as corroboration, we estimate local average treatment effects (LATEs) showing that clean water access reduces self-reported sickness by 23 to 62 percent and leads to reductions in health expenses and missed work due to illness. Our survey evidence also shows that treated households also spend meaningfully less time collecting water.

Our second finding is that households value clean water highly. Because some households in our intervention directly trade off money and clean water, we are able to derive two incentivized valuation metrics from our experiment: a willingness-to-pay (WTP) measure from the prices arm, and a lower-bound measure of willingness-to-accept (WTA) derived from households' decisions to forgo cash payments in lieu of water in the exchangeable entitlement arm. Measuring both is substantively important in our setting.

WTP is the standard measure of valuation, describing the maximum amount of money a household would give up to go from drinking dirty to clean water, and is necessary for a Kaldor-Hicks evaluation of public goods provision absent market failures. However, revealed preference measures of WTP can be hard to interpret because they may be biased downward by various market failures (Greenstone

and Jack 2015). In contrast, WTA describes the minimum amount of money a household would have to receive in order to give up clean water in favor of unsafe water. One way to decide which valuation is more appropriate is to consider property rights: If households have no right to clean water, then WTP may be the relevant metric, but if the government treats access to clean water as a right (as is guaranteed by Article 21 of the Indian constitution), WTA may be the relevant measure of valuation. Even in the absence of market failures or behavioral biases, theory suggests that in cases where a good (in this case, clean water) is both highly valued and has no good substitutes, WTA can (significantly) exceed WTP (Hanemann 1991).

In the literature, direct evidence on the WTP for clean water is scant (Ahuja, Kremer, and Zwane 2010; Null et al. 2012), because prior work has either quantified households' valuation of water treatment technologies rather than clean water itself (Ashraf, Berry, and Shapiro 2010; Kremer et al. 2011a; Berry, Fischer, and Guiteras 2020), inferred valuation indirectly from time costs (Kremer et al. 2011b), or uses contingent valuation (Pattanayak et al. 2005). We estimate a population average WTP from our experimentally measured demand curve for clean water of Rs 132 per month for clean water access (\$20 annually at Rs 80 per US dollar), about 1.5 percent of median household monthly expenditure. The most comparable estimate to our number comes from seminal work by Kremer et al. (2011b), who use travel cost methods to indirectly infer an annual WTP of \$4.44 (adjusted to 2023 dollars) amongst rural Kenyan households for year-long access to a protected (clean) spring. Our estimate is over 4.5 times as high. The WTP we measure for home-delivered clean water is also substantially higher than chlorine-based point-of-use treatment. Adjusted to 2023 dollars, we estimate the population average WTP for chlorine solution from Ashraf, Berry, and Shapiro (2010) at about \$0.11 (14 times lower) for an amount of water equal to the monthly consumption in our sample. Indeed, the WTP from other similar studies is even smaller (Kremer et al. 2011a). These extraordinarily low valuations reflect the substantial gap between the monetary costs of point-of-use chlorine and its take-up. A key implication of our experiment is that this does not represent how much households value clean water itself, in part because prior work has evaluated bundles that provide water coupled with other nonmonetary costs such as travel time, chlorine taste, or the inconvenience of home treatment.

We estimate a lower-bound for WTA that is even larger, at Rs 420 per month (approximately \$60 annually), or 4.7 percent of median expenditures. Our WTA estimates are high enough to exceed the full variable costs of providing water to households for free. Our exchangeable entitlements arm directly points to a new type of contract the government could offer as an adjustment on the large, state-sponsored cash transfers to the poor, which are standard in developing-country settings, including our own (Banerjee et al. 2024; Niehaus and Suri forthcoming). For a component of the full transfer, rather than have the default contract be that cash arrives if households take no action, customers could be offered a default contract where water arrives if households take no action. Our results suggest that many households may choose the water and governments may wish to test this idea in practice.

Does high demand and valuation imply that this approach can contribute to solving the clean water access problem? The answer to this question depends on whether direct delivery of decentralized treated water is amenable to implementation at scale and whether it is cost effective. On implementation, the intervention

appears technically feasible to sustain, given that our partner firm has worked for over a decade in a few hundred villages in Odisha, and demonstrated by the proliferation of private provision of clean water in low- and middle-income countries (Cohen and Ray 2018), with decentralized water kiosk and “water on wheels” delivery models like those used by our implementation partner becoming increasingly common (Advani et al. 2011; Brown et al. 2011; Daly et al. 2021).

Our data suggest that providing clean water at high prices can be privately profitable. However, our estimates predict that at these prices, only a small fraction of households will buy water. These results explain why the private market has not fully solved the universal clean drinking water access problem. Thus, in order to achieve population-wide access to clean water using this approach, the state likely needs to intervene in the market, which is typically justified by the presence of externalities, and judged on the basis of cost per DALY. We carry out a back-of-the-envelope calculation using health benefits based on Kremer et al. (2023), and take-up and cost estimates from our data. We estimate that the cost per DALY of free home delivery of clean water in our intervention ranges between \$71 and \$226, which easily clears conventional cost-effectiveness benchmarks, at costs similar to or slightly higher than well-studied interventions such as coupons or free dispensers for chlorine (\$106 and \$33 per DALY, respectively). This suggests that policymakers interested in expanding clean water access should consider a role for decentralized treatment and home delivery. We discuss these issues in more detail in Section V.

This paper both contributes to the literature on improving access to clean water and joins a broader conversation in the environmental economics literature on measuring how much people in low-income countries value environmental quality (Greenstone and Jack 2015). We demonstrate that decentralized treatment and home delivery of free or subsidized water could play an important role in improving clean water access. In doing so, we also provide revealed-preference experimental estimates of the WTP for an aspect of environmental quality (in this case, water clean enough to drink). Past work has either inferred this type of valuation by using a hedonic approach, valuing the environment via variation in time costs (Kremer et al. 2011b), the cost of air purifiers (Ito and Zhang 2020), or the cost of protective masks (Baylis et al. 2024); or has instead used take-it-or-leave-it, or Becker, DeGroot, and Marschak (1964), methods to estimate demand for environmental quality *production technologies* such as chlorine solution (Ashraf, Berry, and Shapiro 2010) or water filters (Berry, Fischer, and Guiteras 2020), rather than directly for the environmental amenity itself. Because our estimate of WTP for clean water is substantially higher than that found in prior work, this distinction appears meaningful. We also provide amongst the first experimental revealed-preference estimates of WTA for clean water or environmental quality more broadly, extending a literature that has been reliant on contingent valuation (Horowitz and McConnell 2002; Tuncel and Hammitt 2014). Our results suggest that home delivered clean water is highly valued by households and may be an important complement to piped water and point-of-use chlorine.

I. Background

The setting for this paper is the Indian state of Odisha. Odisha is one of the poorer regions in the country, with its HDI ranking twenty-ninth out of 36 Indian states

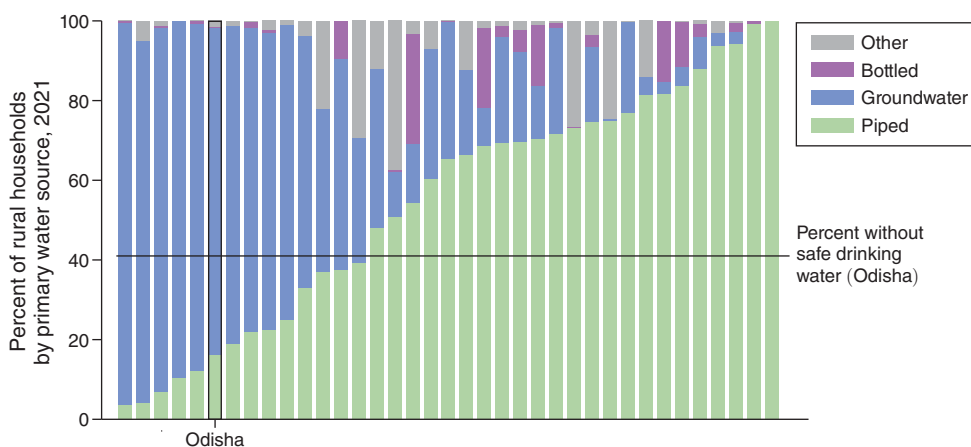


FIGURE 1. WATER ACCESS IN ODISHA

Notes: This figure plots the share of rural households by primary water source (piped, groundwater, bottled water, or other) across all Indian states and union territories in 2021, using data from the seventy-eighth round of the National Sample Survey, with Odisha outlined in black. The black shows the share of households in Odisha found to lack safe drinking water in an independent survey of nearly 10,000 villages in 2023.

Sources: Rukmini S. (2024); National Sample Survey Office (2023); Atmashakti Trust (2023)

and union territories in 2017 and 2018 (MOSPI 2021). As with many parts of the developing world, Odisha faces a severe drinking water access problem, especially in rural areas. Figure 1 plots the share of rural households by primary water source (piped, groundwater, bottled water, and other) across Indian states and union territories per the seventy-eighth round of the National Sample Survey (National Sample Survey Office 2023). With respect to piped water, Odisha ranks thirty-second out of 37, with 83 percent of households still lacking access as of 2021. Moreover, a 2023 survey in 9,856 villages revealed that 41 percent of the population lacked access to safe drinking water (Atmashakti Trust 2023). Drinking water became the subject of grassroots protests (Express News Service 2024b) and was a heated election campaign topic (Express News Service 2024a) in Odisha during India's 2024 general election. Interestingly, privately supplied bottled water, though rare in Odisha, now has a meaningful presence in some wealthier states (Rukmini S. 2024).

Study Population.—We conducted our study in 160 villages spanning six of Odisha's districts; we map the study location in Supplemental Appendix Figure B.1. Our study population was relatively poor, with the control group reporting an average monthly household expenditure of Rs 12,494, or approximately Rs 2,500 in monthly per capita expenditure—meaningfully lower than the contemporaneous all-rural-India average of Rs 3,773 (MOSPI 2024). The median monthly expenditure was Rs 9,000, and median monthly income (elicited using a binned question) was in the Rs 12,000 to 16,000 range. The median household in our sample had five members, of whom three are children. Our sample also had limited (but highly skewed) savings, with the control reporting a median (mean) bank balance of only Rs 2,000 (Rs 22,791). The median household reported being able to make a purchase of no more than Rs 500 ($\approx$ \$6) tomorrow without borrowing.

Clean Water Access.—Although our population was poorer than the national average, our households were not constrained for water. Odisha has an abundance of groundwater at shallow depths and receives substantial rainfall. Averaged across all our surveys over the course of the experiment (August 2022 through August 2023), 76 percent of the control group reported using groundwater for drinking, and 34 percent reported using piped water. Less than 1 percent reported using surface water or bottled water.⁷ Households do report spending time on water collection, with the average household spending 32 minutes a day obtaining drinking water.

Nevertheless, many households lack access to household *potable* drinking water, as river, ground, and surface water contamination are common in Odisha (Odagiri et al. 2016; Senapati 2021; Biswas 2022). Piped water is also not guaranteed to be safe to drink, with prior research having found *E. coli* in piped water in rural areas of the state (Reese 2017), in line with findings from both urban India (Jalan and Somanathan 2008) and other developing countries (World Bank 2018) showing substantial contamination in piped water. Rural piped water in our setting was delivered from storage tanks filled with ground or surface water without central treatment plants, potentially increasing the chance of contamination. As indicative evidence, we collected 17 water samples from different sources (open well, tube well, tap, and the drinking water in our intervention) from our study villages and had them tested by an Odisha state water testing laboratory (see Supplemental Appendix Figure A.2). Though this sample is too small to be representative, we detected problems with pH, *E. coli*, fecal coliform, and salmonella in at least one sample from every source *except* for the treated water from our intervention. We found *E. coli* and fecal coliform in all tap water samples. We did not find evidence of arsenic, lead, or cadmium contamination, consistent with government heavy-metal testing in the region (Ministry of Jal Shakti 2022). There have been several operational challenges with the rural piped water network in Odisha that lead to a risk of biological contamination. These include very little systematic testing of water quality, no residual chlorine in most samples, and village level committees that face difficult management and maintenance challenges (Jal Jeevan Mission 2022a).

Many households appeared to be aware that their water was not safe, and took steps to treat at least some of it. Aggregated over monthly surveys during the experiment, 13 percent of households in the control group reported using chlorine in at least one survey round, 19 percent of households reported boiling water at least once, and 39 percent reported straining at least once. Nevertheless, regular water treatment was somewhat more limited, with only 3.2 percent of households reporting using chlorine, 9.6 percent reporting boiling, and 19.5 percent reporting straining in the average survey round.

Implementation Partner.—In 2022, we began a collaboration with Spring Health India Pvt. Ltd.—a private company that sells clean water to rural households in Odisha. Spring Health was founded in 2011, and in 2022 operated in 230 villages in seven districts of the state. We partnered with Spring Health to conduct a field experiment, discussed in more detail in the next section.

⁷Households could use multiple sources, so the total exceeds 100 percent.

The clean water sold by Spring Health originally comes from local ground water or surface water and is treated in a plant powered using decentralized solar electricity. In most cases, there is one treatment facility per village. Spring Health trains a local entrepreneur who operates and maintains the treatment facility; this entrepreneur is normally also the owner of the well providing the input water. The company pays out a monthly stipend to both the operator of the treatment plant and to its delivery staff. We discuss the various elements and costs of this business model in more detail in Section V as part of a cost-benefit analysis of our intervention.

Spring Health treats its water using an electrochlorination process using solar panels to power the treatment plant. The use of solar panels allows the treatment plant to operate without access to electricity from the grid. Sediments are removed using a precipitation tank and water is disinfected using chlorine dioxide gas delivered in precisely measured doses. This procedure is intended to remove coliform and other bacteria and protozoan parasites. A coagulant flocculant is used to remove fluoride, mercury, arsenic, and iron. Throughout this paper, unless explicitly specified, we use “treated water” or “clean water” to indicate water that has been treated to remove coliform, or that has passed a coliform test.

In Spring Health’s status quo business model, any household in a village served by Spring Health could pay for clean water deliveries to their home. Treated water is packaged and sealed in reusable bottles and delivered by Spring Health delivery staff to the doorstep of enrolled households.⁸ Deliveries are made multiple times each week. Households could place orders as they like during the week, with payments made against orders fulfilled. Consumers had some flexibility with regards to payment timing: 70 percent of consumers pay at the end of the month, 10 percent pay on a daily basis, and 20 percent pay for the full month in advance. Customers who did not pay would not continue to receive water. As the experiment population may be liquidity constrained, this flexible contract helps us approach the WTP for water while mitigating the influence of credit market failures. Prior work points to the importance of flexibility. For example, Tarozzi et al. (2014) find that demand for bednets was higher when payment in installments was allowed.

A key benefit of this model is that it eliminates some of the nonmonetary costs associated with point-of-use treatment or source improvement. First, although treatment is decentralized and, hence, far cheaper than piped water networks, it occurs outside the household. This reduces the inconvenience costs of having to remember to treat and store water at home. Second, it ensures that water is appropriately treated and does not leave an unpleasant residual taste.⁹ Third, the home delivery model eliminates time costs for households that may otherwise have had to travel to collect water, while also removing the risk of contamination in this process. An important remaining risk is contamination at home, underscoring the continued importance of hygiene behaviors and habits such as boiling.

⁸86 percent of our households live in 10 L bottle villages, and the remainder in 20 L bottle villages. These reusable bottles are akin to those used in water coolers, limiting plastic pollution when compared against single-use plastic water bottles.

⁹The use of ClO_2 gas for treatment avoids the well-known unpleasant taste from using chlorine tablets or solutions, which use stable salts such as $NaOCl$, resulting in free chlorine and chloramine compounds containing nitrogen (Crider et al. 2018). In the taste tests we conduct, all participants ranked water treated with chlorine solution (per the package’s specifications) worse than both bottled water from the market (Bisleri brand) and Spring Health water (see Supplemental Appendix Figure A.1).

II. Experimental Design

A. Research Objectives

The design of our experiment is motivated by two main goals. First, we aim to test the viability of home delivery of clean water as a solution to the access problem. Specifically, we seek to measure whether households order clean water, how much households order, how these orders change over time, and how this varies with the price of water. Combining this information with data on costs also reveals whether it is privately profitable to deliver clean water.

Second, we aim to measure whether—and how much—households value clean water. Empirical evidence on whether households value clean water itself, as distinct from water treatment methods, is critically important. If households do not value clean water, whether due to undervaluation of health, lack of information, or other behavioral factors (Dupas and Miguel 2017; Kremer, Rao, and Schilbach 2019), this would explain limited take-up of point-of-use treatment and have important implications for any other approach, including piped water infrastructure. Conversely, if households do value clean water, uncovering solutions that deliver this product outside of previously evaluated approaches may have high returns.

Health benefits that can be perceived by households are likely an important component of how much people value clean water. Another useful outcome is therefore the impact of clean water access on self-reported health. As secondary outcomes, we are also interested in the extent to which home delivery of clean water affects drinking water sources, water collection time, and water treatment. Our experiment is thus designed to facilitate estimation of these features of water demand and subsequent benefits of clean water access.

In our empirical setting, two measures of valuation are of interest: both the amount a household would be willing to pay to obtain clean water (WTP) and the amount a household would have to be paid to forgo clean water access (WTA). WTP is useful as the standard Kaldor-Hicks measure of valuation, and to facilitate comparisons with the (limited) prior literature. However, in a developing-country setting, WTP may be biased downward due to market failures (Greenstone and Jack 2015), while WTA, in theory, does not suffer from this problem. Moreover, WTA and WTP need not coincide: If there are no (or limited) market substitutes for a good, which is the case for clean water, WTA can be much larger than WTP (Hanemann 1991). The WTA is itself also meaningful, as it reveals whether households would prefer receiving free water to receiving cash. This is likely to be policy-relevant in the large number of low- and middle-income countries worldwide which provide low-income households with cash transfers (Banerjee et al. 2024). Moreover, WTA is the correct measure of valuation when the government (as India's does) treats water as a right.

The prior literature has either measured WTP using hedonic approaches (e.g., Kremer et al. 2011b) or instead measured demand for water treatment technologies (Ashraf, Berry, and Shapiro 2010; Kremer et al. 2011a; Dupas et al. 2016; Berry, Fischer, and Guiteras 2020); WTA is typically measured using contingent-valuation approaches (Horowitz and McConnell 2002; Tunçel and Hammitt 2014). Our study is therefore designed to directly measure WTP and WTA for clean water itself, using revealed preference via a randomized controlled trial. Because our

treatment is clean water that is directly delivered to the home, in contrast to prior work, we estimate how much households value clean water absent the disamenities of collection time or water treatment.

B. Sampling

Our experiment took the form of a cluster-randomized field trial with several treatment arms. We selected 160 villages where Spring Health had an existing presence as the site of the experiment. All villages in the experiment had been served by Spring Health for at least 24 months prior to the beginning of the study; the results are, thus, internally valid for these villages but extrapolating outside the sample warrants some caution.

Figure 2 depicts the experimental design. We randomly assigned 120 villages in the sample to one of three treatment arms (with 40 villages per arm), holding 40 villages back to serve as a buffer for necessary replacements/“pure control” group where no experiment activities or survey data collection took place. In treatment villages, we randomized every household in the village to either a (sub-)treatment arm, in the form of special offers from Spring Health (*nominally* 39 households per village, see randomization below), or to control (all other households). In 107 of the 120 main experiment villages, water is sold in 10 L bottles for Rs 1.4 per liter. In the remaining 13 villages, Spring Health sold water in 20 L bottles for Rs 1.25 per liter. Four of these villages were randomly assigned to the price arm, six to the free ration arm, and three to the exchangeable entitlement arm.

Attrition.—Early in the experiment, our partner lost a valuable revenue stream from the sale of carbon credits. They scaled down operations on a village-by-village basis in order to continue to be able to reliably provide clean water. As a result, our experiment could only be implemented in 99 villages (36 in the discount group, 27 in the exchangeable entitlement group, and 36 in the free ration group), rather than the original 120. Supplemental Appendix Figure C.1 shows that our treatment arms are pairwise balanced *after* accounting for this attrition. Moreover, we include village fixed effects in all regressions, ensuring that our identification comes from *within-village* comparisons of households randomly assigned to treatment or control conditions, which are not subject to any village-level attrition concerns.

C. Treatment Arms

Priced Water.—In each village in the “prices” arm, we randomly assigned 13 households to receive a 10 percent discount offer for the duration of the experiment, 13 to receive a 50 percent discount offer, and 13 to receive a 90 percent discount offer. All remaining households received no discount, but were able to continue to order Spring Health water at the full market price.

Free Water Ration.—In each village in the “free ration” arm, we randomly assigned 39 households to receive a free and unconditional ration of up to 400 L of water per month (set to be well above preexperiment average consumption). To receive any water, households needed to place orders with Spring Health, just like

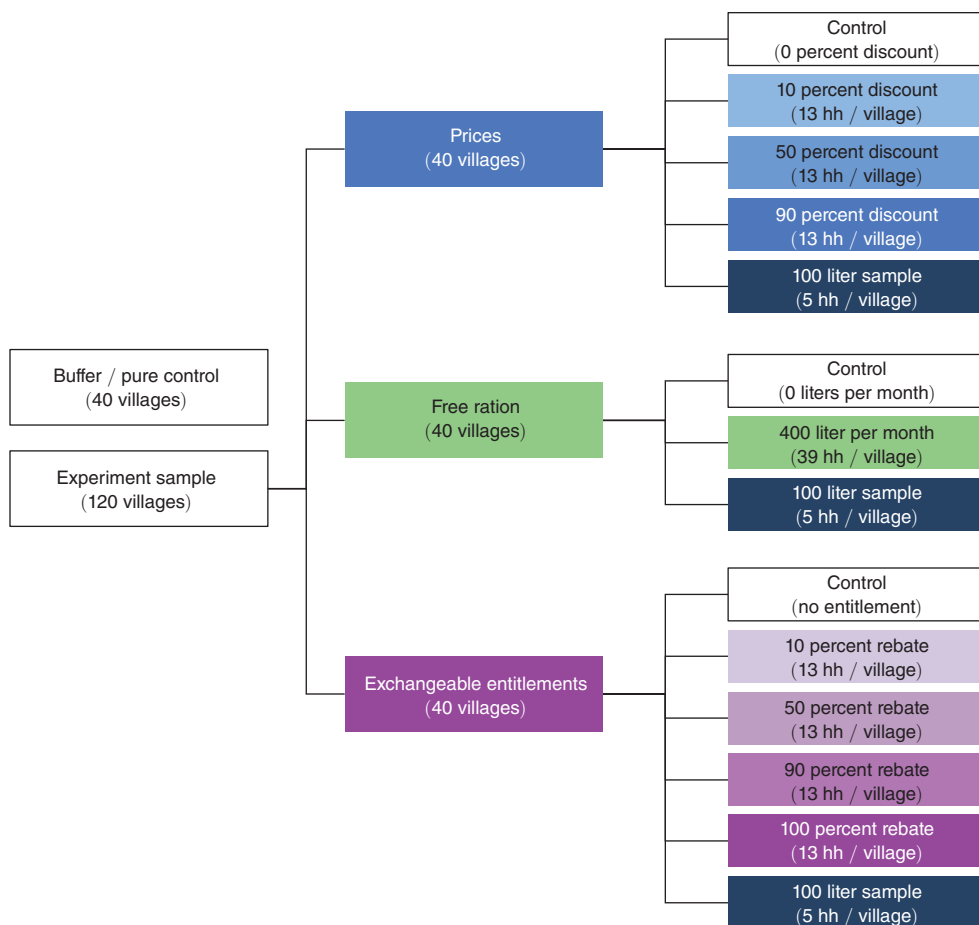


FIGURE 2. EXPERIMENTAL DESIGN DIAGRAM

Notes: This figure shows the experiment design (preattrition). We randomly assigned 40 villages each to a price arm, an exchangeable entitlement arm, a free ration arm, and a buffer/pure control group (no intervention or surveys). In treatment villages, we randomly assigned 39 households to sub-treatments, 5 to a one-time free sample condition, and the remainder to control. In the price arm, treatment households could order clean water with discounts of 10, 50, or 90 percent. In the exchangeable entitlement arm, treatment households could order up to 400 L of water per month for free, and received 10, 50, 90, or 100 percent rebates for any water they chose not to order. Treatment households in the free ration arm could order up to 400 L of water per month for free, but received no rebates.

paying customers. Households that exhausted their ration could order additional water at full price. Households could opt not to use some or all of their quota, since water deliveries were only made when requested. There was no penalty or benefit for households that chose not to avail the full ration. The remaining households received no ration (equivalent to a free quota of 0 L per month), but could continue to order Spring Health water at the full market price.

Exchangeable Water Entitlements.—In each “exchangeable entitlement” village, we randomly assigned 38 households to receive an offer of a 400 L entitlement, just as in the free ration condition. As above, in order to obtain this water, they needed to place an order with Spring Health. However, in this arm, households could redeem

unused water within their entitlement amount for cash. For every unclaimed bottle of water, households were randomly entitled to receive payments equal to 10 percent of the market price (nine households), 50 percent of the market price (ten households), 90 percent of the market price (nine households), or 100 percent of the market price (ten households).

The reimbursement rates were set exactly equal to the prices in the prices arm, creating similar monetary incentives differentiated only by whether ordering an additional bottle involved paying out versus forgoing cash. At no point did this condition involve any physical exchange of water for cash. As in the other arms, households that did not receive a nonzero exchangeable quota remained eligible to buy Spring Health water at full price.

Since the entitlement was monthly, each household's refund was calculated at the end of every month. Transfers were made either to mobile money accounts or bank accounts using details provided by households. Though households were intended to be paid at a monthly frequency, in practice, payments were made less often, in part because refund amounts for a single month were often very small. Supplemental Appendix Figure C.3 demonstrates that households do not change water ordering behavior in response to payments, suggesting that payment timing does not impact how households order water in this arm. In order to confirm that households understood this treatment, we conducted a small set of phone-based spot checks, which revealed high comprehension. Of the 28 households across four villages we reached for this survey, representing all price groups, when asked how much cash they would receive if they did not order one bottle, 24 of these households named the correct amount, 2 households were off by Rs 1, 1 household did not know, and 1 household refused to answer.

All Treatment Villages: Free One-Time 100 L Sample.—In addition to the treatments described above, we randomly assigned five households in each treatment village to receive a one-time offer of 100 L of free water. Households that chose not to take this offer received nothing in return. These households could otherwise order water at the market price. This condition provides a test of the extent to which experience with clean water impacts demand.

Buffer/Pure Control (40 Villages).—We assigned a final 40 villages to a buffer/pure control group, from which we drew replacement villages in the event of main-sample attrition. The remaining villages in this arm were completely “business as usual.” Throughout the experiment, all households in these villages were able to order Spring Health water just as they had been doing before, but no surveys were conducted in these villages. Our only source of data on these villages comes from Spring Health's administrative data, generated through the normal course of business. We compare households in these villages to control households in treated villages in Supplemental Appendix E.

D. Randomization

We intended to include all households in each treatment village in our study, and, therefore, conducted an in situ randomization procedure, in which we distributed

scratch cards drawn from a shuffled deck to every home in the village. We prepared decks based on the most recent census figures on village population, and each village's deck included 39 treatment cards. Since the number of households in most villages had changed since the census and since not every household had an available adult to receive a card during distribution, decks were not exhausted. Thus, in some villages, the actual number of treatment cards distributed was less than 39. The average number of realized treatment households per village was 36.25. Supplemental Appendix Figure B.2 shows the scratch card design. All scratch cards looked identical, but had a unique ID number that the research team could use to map scratch cards to treatment conditions. This mapping was unknown to the field staff; offers were only revealed when a household scratched their card in front of the enumerator. The household address and a mobile phone number was noted down when the card was distributed. This helped ensure that cards could not be used by anyone other than the recipient households.

E. Treatment Duration

The experiment began in May 2022 and concluded in August 2023. For logistical reasons, villages were randomly assigned to eight phases, and the offers were rolled out in a staggered manner, with village randomization stratified by phase. All treatment households were initially told that their offers would last for five months, based on available funding at the beginning of the experiment. However, we obtained additional funding during the course of the experiment, and were, thus, able to extend the offer for two months in all except the first two waves, which contained 35 villages. Households were informed about their extensions in December 2022–January 2023. We see no change in consumption at the five-month mark (see Figure 4). Supplemental Appendix Figure B.3 shows the implementation timeline, including the scratch card distribution, water distribution, and survey data collection (described in more detail in Section III below).

F. Experiment Integrity

Balance.—Supplemental Appendix Figure C.1 provides pairwise balance tests between each treatment arm and the control group on a series of household characteristics. Due to logistical constraints, we were unable to conduct surveys prior to the start of water distribution. We therefore test for balance using data from the endline survey on variables that were impossible or very unlikely to change as a result of our experiment: household size, presence of children in the household, the education of the household head, years the household head has lived in the village, and ownership of expensive appliances. Because this balance test is conducted on data collected at endline, it accounts for attrition by construction. We fail to reject balance on all variables and across all pairwise comparisons, and the magnitudes of any differences are small.

Compliance.—Supplemental Appendix Figure C.2 shows that while most of our 99 experiment villages received water deliveries for 100 percent of experiment

months, in 12.3 percent of *village months*, our implementation partner did not offer delivery, largely due to staffing issues. In such months, water orders are necessarily zero for all households. In Section IV, we present the effects of our treatments in the 87.7 percent of village months *where delivery occurred*. Our primary empirical object of interest is households' valuation of clean water, which cannot be evaluated in months where no water was offered. When conducting our cost-effectiveness analysis in Section V, we adjust our compliance measure downward (costs upward) to allow for imperfect reliability.

Preanalysis Plan.—This study was preregistered through the AEA RCT Registry as AEARCTR-0010545 (Burlig, Jina, and Sudarshan 2023). In the preanalysis plan, we list the primary outcomes as follows: (i) household-level treatment effects on water consumption; (ii) price elasticity of demand; and (iii) household-level treatment effects on health.¹⁰ We list nonhealth benefits and spillovers as secondary, and also describe heterogeneity and multiple hypothesis testing. We generally followed this plan. We report the minor deviations from the preanalysis plan in detail in Supplemental Appendix H.

III. Data Collection and Outcome Variables

A. Record of Scratch-Card Distribution

During our visits to all households in each treatment village to distribute scratch cards, we generated a “listing” dataset, which includes the village name, the name of the household head, whether the household was a Spring Health customer prior to the experiment, contact and address details, the offer type, and the scratch card ID, which allows us to confirm the link between a household and its treatment offer. We use these data to define the universe of households in each of our treatment villages.

B. Administrative Data

Our main outcomes of interest concern clean water demand. We obtained administrative data on water orders from Spring Health. For every household that orders clean water—including both households that received a treatment offer and those that did not—we observe daily information on the number of bottles of water they ordered from Spring Health, and at what price. For exchangeable entitlement households, we use these data to calculate how much money they are owed at the end of each month.

Because Spring Health's administrative data are complete (i.e., they contain entries for every single bottle of water ordered in each village), and our listing dataset enumerates every household in each village, we can also infer that households that do not appear in the Spring Health administrative dataset must consume 0 L. This yields a household-by-date panel of Spring Health water orders.

¹⁰Though not included in the PAP, our ex ante power calculations, based on water consumption data from our pilot and health data from Clasen et al. (2015), suggested that we should be powered to detect a demand elasticity of -0.03 , water consumption differences of 5 percent or greater, and health effects of 20 percent–30 percent.

C. Survey Data

We administered a series of surveys with a randomly selected subsample of 13 households in each treatment village, stratified across sub-treatments to maximize statistical power.¹¹ We visited each survey household for four short “high-frequency” checks and a longer endline. The first high-frequency survey normally occurred in the first or second months of the treatment start date, while the endline normally occurred in the last month.¹² Where possible, we repeatedly surveyed the same households in each survey, except in cases where a household dropped out. In such cases they were replaced by selecting a backup household at random from the corresponding (sub-)treatment arm.

In each survey, we asked households about drinking water choices, water treatment choices, and health in the week before the survey. Our health outcomes include health expenses, missed work due to illness, self-reported incidents of perceived sickness, and a variety of self-reported symptoms of waterborne disease: vomiting, fever, stomach ailments, flu symptoms, and other symptoms, all of which have been linked to drinking contaminated water (World Health Organization 2017). To keep the length of repeated surveys manageable, we did not collect placebo health outcomes (e.g., broken bones) and our results. Because these are self-reported measures, they can help explain why households value water, but they cannot be interpreted as clinical outcomes.

In the first survey, we additionally collected basic household demographic information, including total number of household members, number of adults/children, monthly income and occupation. In the endline survey, we also collect more data on households including information on household savings, expenditures, liquidity, and asset ownership. We did not conduct surveys during months with no water delivery, so we are unable to evaluate differences between treatment and control during these periods.

D. Water Testing

As discussed in the background section, we conduct two types of water tests on small samples. First, we collected 19 water samples from households in our study villages (8 from open wells, 6 from tube wells, 4 from taps, and 1 Spring Health) and had water quality (including pH, heavy metals, and biological contaminants) analyzed by a government laboratory. Supplemental Appendix Figure A.2 shows the results. While this is not statistically conclusive, we find substantial biological contamination among all but the Spring Health sample and no evidence of heavy metal issues.

¹¹ In price discount villages, we selected four control households and three households in each discount level for surveys; in pure quota villages, we surveyed six control households and seven quota households; and in exchangeable quota villages, we surveyed three control households, two 10 percent exchange households, three 50 percent exchange households, two 90 percent exchange households, and three 100 percent exchange households. We did not survey in pure control villages.

¹² An exception are the 35 villages in the first two phases where the offer duration was shorter. Here, the last survey occurred three months after the offers ended. Supplemental Appendix Figure B.3 shows survey timings by calendar month for all phases.

Second, we conducted nine water taste tests with members of our survey enumeration team. In these tests, subjects were given a comparison cup of water (unbeknownst to the subjects, this was Bisleri, a leading bottled water brand), and then asked to compare four additional samples (provided blind, and in a randomized order) to this original sample. These samples were as follows: tap water plus added chlorine (per the instructions on the chlorine treatment packet), Bisleri plus added chlorine, the treated water in our treatment, and treatment water plus added chlorine. Supplemental Appendix Figure A.1 shows the results. Our taste testers uniformly rate chlorinated sources the lowest and rank Spring Health the highest on average.

IV. Analysis and Results

In this section, we describe the specifications we use to analyze data from the experiment and summarise the main results. We begin by describing patterns of demand for clean water under our different treatment arms. Next, we turn to our results on the valuation of water. Finally, we provide survey evidence on the effects of water offers on time use, water treatment behaviors, and health. Since households in study villages were familiar with Spring Health and its product, our results can be interpreted as the impact of water offers on demand, without also introducing new information about a new product.

A. Demand for Clean Water

We use administrative data on water orders to estimate demand under all of our treatment arms. Households see nonzero marginal costs under both the prices and exchangeable entitlements arms. In the first case, ordering water requires households to pay a per unit price, and in the second, ordering water requires households to forgo an equivalent per unit cash transfer. In the free ration arm, the marginal cost is zero.

The top panel of Figure 3 shows the demand curve, plotting average water orders at each price level for the priced water arm in blue, and at each refund level for the exchangeable entitlement arm in purple. The green point at a zero price reports the average water orders for households in the free ration arm.¹³ The bottom panel of Figure 3 separates net demand into changes on the extensive margin on the left (probability of a household ordering *any* water in a month) and the intensive margin on the right (conditional on ordering, how *much* do households order).

We also describe changes in demand relative to the status-quo levels in the control by estimating a set of simple regression models, described in equation (1), and report the results in Table 1.¹⁴

¹³For graphical clarity in Figure 3, we do not use data from the 20 L bottle villages where prices are slightly lower. These villages are included in Table 1.

¹⁴In estimating equation (1), we restrict the estimation sample to exclude months where water delivery was disrupted and households could not place orders. See Supplemental Appendix C for more details on these interruptions.

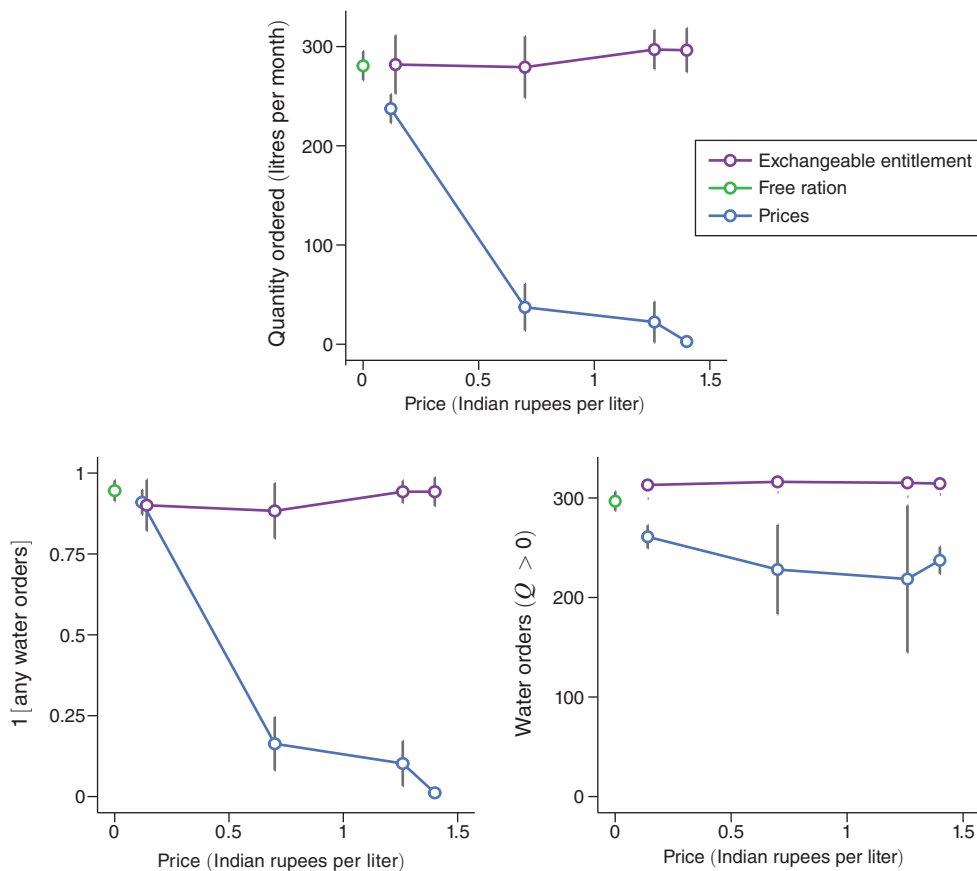


FIGURE 3. DEMAND FOR CLEAN WATER

Notes: This figure presents the demand curve for clean water. We plot monthly water orders against the price of water for households in the price arm and control (i.e., full price) in blue and against the refund amount for the exchangeable entitlement arm in purple. The green point at 0 price shows mean orders in the free ration arm. The top panel plots average water orders at different prices (or refund rates). The bottom-left panel plots the probability of ordering any water. The bottom-right panel plots quantity ordered conditional on ordering water. We show 95 percent confidence intervals, derived from standard errors clustered at the village level, in light gray. We jitter the Rs 0.14 price point slightly to the left for visual clarity. In the experiment, both the price arm and the exchangeable entitlement arm included an identical Rs 0.14 incentive level.

$$\begin{aligned}
 (1) \quad Y_{it} = & \eta_1 \cdot 10\% \text{ discount}_i + \eta_2 \cdot 50\% \text{ discount}_i + \eta_3 \cdot 90\% \text{ discount}_i \\
 & + \eta_4 \cdot 10\% \text{ exchange}_i + \eta_5 \cdot 50\% \text{ exchange}_i + \eta_6 \cdot 90\% \text{ exchange}_i \\
 & + \eta_7 \cdot 100\% \text{ exchange}_i + \eta_8 \cdot \text{Free ration}_i + \eta_9 \cdot \text{One free}_i + \gamma_v + \theta_t + \varepsilon_{it},
 \end{aligned}$$

where the outcome Y_{it} is either an indicator equal to 1 if the household i ordered any water during the course of the month-of-sample t , or the total monthly water orders for household i in month-of-sample t in liters, Q_{it} . The named dependent variables correspond to the treatment arms, γ_v and θ_t are village and month-of-sample fixed

TABLE 1—INTENT-TO-TREAT EFFECTS OF CLEAN WATER OFFERS ON WATER ORDERS

	Any orders		Orders in liters	
	(1)	(2)	(3)	(4)
Prices (discounts)	0.38 (0.02)		95.84 (7.68)	
Exchangeable entitlement	0.90 (0.02)		290.65 (11.17)	
Free ration	0.89 (0.02)	0.89 (0.02)	269.89 (6.97)	269.89 (6.97)
Onetime 100 L	0.14 (0.01)	0.14 (0.01)	13.42 (1.96)	13.41 (1.96)
10 percent discount		0.09 (0.03)		20.24 (9.33)
50 percent discount		0.15 (0.04)		34.08 (10.84)
90 percent discount		0.88 (0.02)		232.12 (7.22)
10 percent rebate		0.89 (0.04)		284.69 (14.30)
50 percent rebate		0.87 (0.04)		282.34 (14.76)
90 percent rebate		0.93 (0.02)		298.58 (10.24)
100 percent rebate		0.93 (0.02)		297.53 (11.31)
Control means	0.012	0.012	2.818	2.818
Prices = exchange	< 0.01		< 0.01	
Prices = ration	< 0.01		< 0.01	
Exchange = ration	0.689		0.115	

Notes: This table presents intent-to-treat effects of water offers on water orders at the monthly level, estimated using equation (1) or a pooled version thereof. We restrict the sample to village-months where Spring Health delivered water. In columns 1 and 2, the outcome is a binary indicator for the household having ever bought water during the month. In columns 3 and 4, the outcome is total water orders in liters per month. All regressions include village fixed effects. Standard errors are clustered by village. Price = exchange, Ration = exchange, and Price = ration are tests for equality between the impacts of the respective treatment groups on average monthly orders, pooled across discount and exchange levels.

effects, and ε_{it} is an error term. Identification comes from a comparison of treated consumers to untreated customers *within* villages.

Finally, we estimate a dynamic specification to study the effects of our treatments on water orders over time, with results reported in Figure 4. We pool sub-treatment arms for clarity and report coefficients for a fully disaggregated variant in Supplemental Appendix Figure D.1.

$$\begin{aligned}
 (2) \quad Y_{it} = & \sum_{r=1}^7 \beta_1^r \cdot \text{Any discount}_i \times \mathbf{1}[\text{offer month} = r]_{it} \\
 & + \beta_2^r \cdot \text{Any exchange}_i \times \mathbf{1}[\text{offer month} = r]_{it} \\
 & + \beta_3^r \cdot \text{Free ration}_i \times \mathbf{1}[\text{offer month} = r]_{it} \\
 & + \beta_4^r \cdot \text{One free}_i \times \mathbf{1}[\text{offer month} = r]_{it} + \gamma_v + \delta_t + \varepsilon_{it},
 \end{aligned}$$

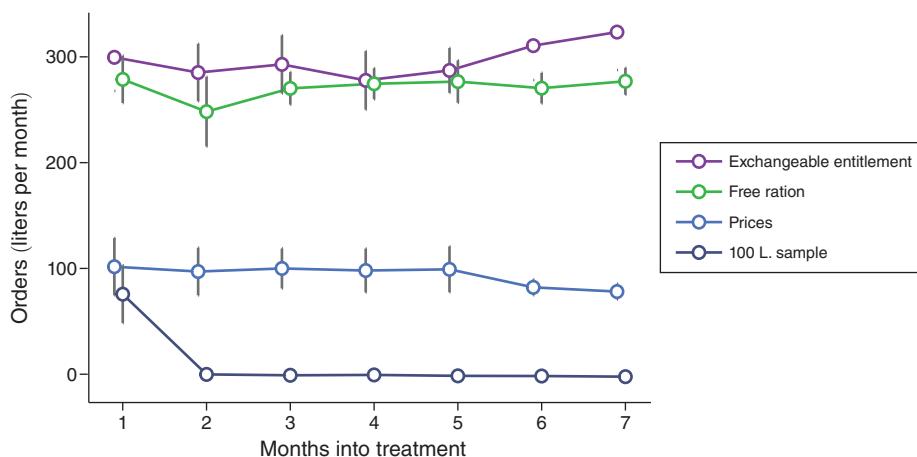


FIGURE 4. WATER ORDERS OVER TIME

Notes: This figure plots the effect of our treatments on water orders as a function of time since treatment started (with the first month offers were active set to 1), estimated using equation (2). We pool over all price and exchangeable entitlement sub-treatments. Standard errors are clustered at the village level. 35 villages—the first enrolled in the experiment—had only five months of treatment, while the remainder had seven. The sample is restricted to months when deliveries occurred. Price and exchangeable entitlement points are jiggered to the left for visual clarity.

where $\mathbf{1}[\text{offer month} = r]_{it}$ is an indicator for being $r \in \{1, 7\}$ months into the treatment offer (such that 1 is the first month of the offer), and all other terms and sample restrictions are as above.

Access to Clean Water.—While water purchases are low in the control (i.e., full-price) group, our intervention leads to very high clean water take-up. We can reject impacts on take-up smaller than 34 percentage points (prices), 86 percentage points (entitlements), and 85 percentage points (ration). The bottom left panel of Figure 3 shows that the monthly probability of ordering any clean water rises as price falls, reaching approximately 90 percent at a price of Rs 0.14 per liter (90 percent discounts). Accordingly, we strongly reject equality between the 10 percent discount and 90 percent discount effects (p -value < 0.01).

Importantly, we see no evidence of sharp changes in demand around a price point of zero (a so-called “zero price effect”). Indeed, we cannot reject equality of take-up on the extensive margin at 90 percent discounts and zero price ration ($p = 0.763$) and observe a moderate and statistically significant increase in average orders (232 L versus 270 L, $p < 0.01$). These patterns are in stark contrast to prior work on chlorine-based water treatments, where take-up *even at zero price* is relatively low (Kremer et al. 2023).

However, we find that households are quite price sensitive. As shown in column 2 of Table 1, moving households from the original market price to a 90 percent discount increases the probability of a monthly orders by 88 percentage points (SE 0.02). Notwithstanding the variation in order probability with price, an interesting feature of demand is that *conditional* on consumption, quantity demanded is relatively inelastic. The bottom-right panel of Figure 3 shows demand along the intensive margin. In the prices arm, conditional on ordering a nonzero amount of water,

there is no statistically significant variation in orders with price. On the intensive margin, all treatment arms order substantial amount of clean water. Even at the highest price in the sample (Rs 1.4 per liter, ten times the lowest price), mean orders for households that consume more than zero is about 237 L per month. In our data, the average household size in our experiment is five. Consuming 1.5 (2) L per person per day—a standard biological benchmark (EFSA panel on Dietetic Products, Nutrition, and Allergies 2010)—would lead the average household to use 225 (300) L of water per month. The top panel of Figure 3 shows a full demand curve, combining the extensive and intensive margins, which shows quantity decreasing as price increases, suggesting that while this approach can achieve universal access at low prices, higher prices strongly discourage adoption. As the two figures in the bottom panel show, this is driven almost entirely by the extensive margin.

Selection into Drinking Water Purchases.—This pattern of demand can be rationalized by a simple stylized model where we assume that the main benefit to treated water—for households that already have access to *nonpotable* water—comes from improved health. Let a household's probability of falling sick in a month be proportional to the share θ of their biological (i.e., fixed) drinking water needs W that are met by clean water. For instance, let $P(\text{Healthy}) = k\theta$ so that the probability of being healthy conditional on setting $\theta = 1$ (i.e., drinking only clean water) is $k \in [0, 1]$. Typically, we would expect $k < 1$ because, even treated water may get recontaminated at home, perhaps due to poor sanitation practices. Assume further that the cost of falling sick is p_{sick} (e.g., lost work, increased health expenses, etc.) and the cost of purchasing water is p_{water} . In Supplemental Appendix Table D.3 and Table 3, we provide survey evidence on health outcomes that are consistent with these mechanisms.

Households can thus be modeled as maximizing consumption utility $u_i(c)$ subject to a budget constraint $p_c \cdot c + p_{\text{water}} \cdot \theta W \leq I - p_{\text{sick}} \cdot k_i \theta$, where the right hand side is monthly income I less sickness costs. If the product $p_{\text{sick}} k_i \geq p_{\text{water}} W$, households will set $\theta = 1$; otherwise $\theta = 0$.

Lowering the price of water changes the extensive margin and attracts households with lower expected health benefits (i.e., lower values of $p_{\text{sick}} \times k_i$) into buying. This may be households with worse sanitation practices, lower wages (hence lower absenteeism costs), or clean alternatives.¹⁵ There is some weak evidence of this type of selection in Table 3 where we document point estimates of clean water on sickness that are greater for the prices arm than free rations. Admittedly, this model is sparse and selection could also be induced by allowing for factors such as switching costs, which may vary for other reasons.

Under all treatment arms and all prices, mean consumption of clean water (conditional on ordering) is approximately enough to cover households' biological drinking water needs, but not to displace all other water uses. This is consistent with households setting θ near 1. Survey evidence in Supplemental Appendix Table D.2 shows that our treatment offers increase the number of water sources households report using, increase the share of households drinking any clean water, and increase the share of households that report drinking *only* clean water. Taken together, these

¹⁵The fact that better sanitation increases water purchases in this model suggests that households that have access to disinfectants like chlorine but do not like the taste might have a *greater* propensity to pay for home delivered water.

pieces of evidence are consistent with households buying water in sufficient quantities to make it their primary drinking water source, but not to use purchased water for other end uses.

In the model we have outlined, there is no benefit to ordering more clean water than is required to meet drinking water requirements W . In the experiment, households that are provided a 400 L ration of free water do not use it in full. Table 1 shows that take-up under the free ration is 90 percent, and this treatment raises average consumption by 270 L per month per household. This treatment effect translates to a population average consumption of 280 L per month, or 300 L per month conditional on nonzero orders. These results likely reflect the fact that although our households have limited access to clean drinking water, they do not suffer from water shortages in general. Indeed in our survey data, in both treatment and control arms, households report ever running out of water (for any end use) in the prior week only 2 percent of the time.

Modeling water as providing mainly health benefits would be unreasonable if treated households could easily sell or otherwise provide water to people in the control group. To check this, we asked households about reselling in the endline survey, and not a single respondent reported doing so. This is perhaps unsurprising in the context of clean water, where it is difficult to signal quality if it is not delivered by a trusted source in a sealed container, and where water is heavy and difficult to transport.¹⁶

Finally, it is notable that demand is sustained over time. Figure 4 plots dynamic effects of water orders for each treatment arm, estimated using equation (2). Water orders in our main treatments are stable during the full experimental period. These results demonstrate that households are consistently willing to spend (or give up) money for water. This again contrasts with prior experiences using other approaches to water treatment (Ashraf, Berry, and Shapiro 2010; Dupas et al. 2016; Berry, Fischer, and Guiteras 2020), where many households initially took up either chlorine or water filtration but were not found to be using it in a follow-up measurement. Households in our main treatment arms consume similar amounts of water every month, suggesting there are no learning or “experience good” factors at play (either about the intervention itself or about clean water) or “experience good” effects. A direct test of experience good effects comes from households given a one-time free allocation of 100 L. These consumers use this water in the first month, but revert to behaving like the control afterward. Moreover, despite being informed that the intervention would end after several months, households also do not significantly change behavior in the final month of the experiment, suggesting that the temporary nature of the intervention is unlikely to bias our findings relative to a permanent policy.

B. The Value of Clean Drinking Water

Our experiment recovers two measures of valuation: (i) a WTP estimate from the demand curve in the prices arm, and (ii) a bound on a WTA estimate from the exchangeable entitlement arm.

¹⁶In theory, a reseller could hand over a Spring Health bottle to someone else without breaking the seal. In practice, this would render sales unprofitable, since our partner levied a substantial charge of Rs 400 if water containers were not returned by the purchasing consumer.

Willingness-to-Pay.—A measure of the population average WTP is the area under the demand curve in the top panel of Figure 3, which is also then the consumer surplus at zero price. To estimate surplus from drinking water, we calculate the area under the prices (blue) demand curve from Figure 3, imputing linear demand between price points. This yields an average WTP of about Rs 132 for monthly clean water access—approximately 280 L per month, where consumption at the zero price point is given by the average water orders in the free ration treatment arm. These WTP results are robust to alternative functional forms. Fitting a cubic spline through our price points yields a mean WTP estimate of Rs 126. Using a step function with the extreme assumption that all households between price points have the WTP of the adjacent but higher (lower) step yields a mean WTP of Rs 196 (68). At approximately 1.5 percent of median consumption, our central WTP estimate is relatively high for a very poor population (comparable to monthly spending on milk, Rs 186).

Willingness-to-Accept.—In the exchangeable entitlement arm, households have an entitlement of 400 L of water. They may place orders for water for free, but since unused water earns a rebate, this arm provides household the option of relinquishing the entitlement to water in exchange for a varying monetary incentive. Table 1, column 4, shows that assignment to all rebate levels in this arm raises consumption by 280 L to 300 L per month. At the highest rebate rate (Rs 1.4 cash for every liter not ordered) households continue to order a substantial amount of water (roughly 300 L on average), and in doing so, forgo on average Rs 420 ($\sim$ \$5.25) per month to “cover their drinking water needs.”¹⁷ This amount is a lower bound on WTA. If cash rebates had been large enough to induce households to decline water for cash, we would identify WTA precisely, but we do not observe this in our experiment. We cannot reject equality of water orders in liters between 10 percent and 100 percent rebate levels ($p = 0.343$). One concern is that households may value temporary clean water access differently from permanent clean water access. If anything, this should make households in the WTA arm more likely to take the cash, further suggesting that our WTA estimates represent a lower bound.

This is a substantial sum of money, even as a lower bound. Rs 420 is about 4.7 percent of median monthly expenditure in the control group (Rs 9,000), 77 percent of average expenditure on tobacco and alcohol (Rs 598), 2.3 times monthly expenditure on milk, and 84 percent of spending on mobile bills (Rs 498). It is also sufficient to cover the variable costs of providing water for free (see Section V for a discussion of the costs of supply). In Section V, we argue that these measures of WTA have direct policy implications.

Comparing WTP and WTA.—Divergences between measures of WTP and WTA are common (Horowitz and McConnell 2002; Tuncel and Hammitt 2014), but our experimental design and results help rule out certain explanations. First, households in both the price and exchangeable entitlement arm face incentivized choices, ruling out issues with stated preference approaches. Second, our measures of WTP

¹⁷This shorthand is convenient since household orders suggest they do not value the four-hundredth unit of water much but do value the one-hundredth unit, the former being well above the probable drinking water needs of most households and the latter well below.

and WTA are derived in a long-running field experiment, which avoids many lab framing issues (e.g., those discussed in Plott and Zeiler 2005). Third, differences in opportunity costs cannot explain the wedge because the marginal incentives in the price and entitlement arms are identical. Fourth, selection concerns are mitigated by randomized assignment, as the balance tests in Supplemental Appendix Figure C.1 help confirm. Fifth, in both of these arms, households must call the company and ask for a delivery to get water. Households therefore do not have a physical stock of water that they are being asked to return for cash, so differences are unlikely to be due to loss aversion (e.g., Ericson and Fuster 2014). Sixth, this gap is not driven by household mistrust in the exchangeable entitlement rebates. If households in this arm did not expect to be paid, we would expect that the pattern of household orders shortly *after* the first payment was deposited would differ from those observed shortly before. Supplemental Appendix Figure C.3 shows that this is not the case. Seventh, the gap between WTP and WTA is unlikely to be driven by information frictions, as Figure 4 shows that demand is stable over time, and that experience with clean water (in the one-time offer group) does not lead to sustained orders absent discounts. Eighth, we think income effects are unlikely to drive the gap between WTP and WTA, because the overall magnitude of the implied transfer is both very small compared to household income and available only in the future. Finally, households in our sample have limited ability to borrow; prior research demonstrates that liquidity constraints can lower WTP (e.g., Berkouwer and Dean 2022). However, because these constraints increase the (shadow) value of cash, they should impact both WTP and WTA, rather than driving a wedge between the two measures of valuation. Nevertheless, if households do not commonly hold liquid cash, purchasing may be more difficult than giving up rebates.

What, then, explains the difference between WTP and WTA in our setting? Theory suggests one possible explanation: Under certain conditions, WTA can far exceed WTP. In particular, this can arise for goods with limited market substitutes (Hanemann 1991, 2003) or when consumers have asymptotically bounded utility functions (Amiran and Hagen 2003). The intuition is as follows: If there is no bundle of private goods that can compensate for the loss of a particular rationed good in utility terms, then WTA for the rationed good can be infinite. Thus, when there are only imperfect substitutes for the rationed good available on the market, WTA can be much larger than WTP. Support for this hypothesis has been found in lab experiments (Shogren et al. 1994; Shogren and Hayes 1997). This describes our setting well: Given that *clean* (and perhaps also tasteless) water is the commodity of interest, households have only the imperfect substitute of point-of-use treatment, which—by revealed preference—is relatively unpopular. Other commercial bottled water is available, but extremely expensive (e.g., Bisleri-branded water costs Rs 4.5 per liter—more than three times the unsubsidized Spring Health price). Though this need not be the only explanation for the divergence between WTP and WTA—there may be behavioral factors that also drive a wedge between the two measures, such as households treating the entitlement as a labeled transfer—it appears plausible in this context. In Section IVE, we find suggestive evidence of higher WTP (and thus, a smaller gap between WTA and WTP) for higher-income households and those with existing access to piped water.

Households Value Clean Water Highly.—Both our WTP and WTA estimates reveal that households would be willing to exchange substantial sums of money for clean water access, with an annual WTP of approximately \$20, and an annual WTA of approximately \$60. To contextualize our measures, this WTP (WTA) for clean water access is approximately 4.5 times (14 times) larger than the WTP for clean spring water indirectly inferred from time use in Kremer et al. (2011b). It is also much higher than for chlorine-based point-of-use treatment. Adjusted to 2023 dollars, the population average WTP for clorin (a relatively popular brand of home-use chlorine solution), from the demand curve reported in Ashraf, Berry, and Shapiro (2010), is about \$0.11, or 14 times lower than our WTP, for an amount of water equal to the monthly consumption in our sample. An important advantage of our estimates is that they directly reveal households' valuation of clean water itself, excluding any disamenity value from water purification or traveling to retrieve water.

As a comparison with other environmental quality valuations in the developing world, the WTP of households for clean water in our experiment is about 15 times higher (45 times for WTA) than the WTP of low-income Delhi residents to reduce PM_{2.5} by 10 micrograms per cubic meter from Baylis et al. (2024). It is slightly less than double the estimate in Ito and Zhang (2020) for how much richer Chinese households would be willing to pay to eliminate pollution generated by the Huai River heating policy. We therefore conclude that household valuation of clean water is relatively high.

C. Impacts on Water Collection and Water Purification

In this section, we use survey data to evaluate the extent to which clean water leads to household behavior changes. We focus on two categories of outcomes. First, we measure time spent collecting water. The water we deliver in our intervention might lead to substantial time savings that households could find valuable if it were to displace trips to wells to collect groundwater. This is plausible in our setting—34 percent of the control group reports using taps as a source of drinking water and only 24 percent as the sole source.

Second, we measure behaviors households undertake in order to make their water safe to drink. Theoretically, one benefit of purchasing clean drinking water is that home treatment could be reduced, saving time and money, especially for boiling where fuel costs may be significant. On the other hand, given that in-home contamination remains a plausible concern (Jeuland et al. 2021), and that mixed drinking sources may be used, home treatment is probably a very useful habit to retain.

We use data from our repeated surveys to estimate panel regressions of outcomes on our treatments. In the interests of parsimony and precision, in our main specification, we pool across subtreatment arms.¹⁸ Our main specification is, thus,

$$(3) \quad Y_{it} = \eta_1 \cdot \text{Any discount}_i + \eta_2 \cdot \text{Any exchange}_i \\ + \eta_3 \cdot \text{Free ration}_i + \gamma_v + \delta_t + \varepsilon_{it},$$

¹⁸ We present intent-to-treat effects on health broken out by subtreatment in Supplemental Appendix Table D.1.

where Y_{it} is an outcome for household i surveyed in month-of-sample t ; *Any discount_i*, *Any exchange_i*, and *Free ration_i* are treatment indicators; γ_v and δ_t are village and month-of-sample fixed effects, respectively; and ε_{it} is an error term, clustered by village.

Water Collection.—We first consider changes in time spent collecting water. Our treatment offers are for delivered drinking water, so we expect water collection time to fall. The first column of Table 2 reports the results. Control households spend an average of 32 minutes per day collecting water. All three offer types lead to meaningful reductions in water collection time, with time savings of 4.8 minutes (15 percent, $p < 0.01$) in the price arm, 9.0 minutes (28 percent, $p < 0.01$) in the exchangeable entitlement arm, and 12.6 minutes (39 percent, $p < 0.01$) in the free ration arm.¹⁹ These impacts may represent a meaningful benefit to households, helping to rationalize their relatively high valuation of clean water. Note that water collection time does not fall to zero, consistent with households continuing to use external water sources to meet their (nondrinking) water needs.

Water Purification.—Next, we measure the effect of our offers on actions households undertake to make their water safe to drink. Columns 2–4 of Table 2 report effects on the probability that households report boiling, chlorinating, or straining water in the past week.²⁰ We find no evidence that households change their self-reported treatment behavior and can reject relatively modest effects in all treatment arms, suggesting that clean water offers do not crowd out (limited) ongoing water treatment efforts.²¹ Columns 5 and 6 report changes in the collection time and costs of the main fuel used to boil water, conditional on reporting any boiling in that week. The point estimates suggest small-to-moderate reductions in both across all treatment arms, but they are imprecisely estimated.

D. Effects of Clean Water on Self-Reported Health

We use our survey to measure the impacts of access to Spring Health water on a series of health outcomes as reported by households. Prior evidence demonstrates that drinking clean water improves health (Kremer et al. 2023). We find that the water improves several self-reported health outcomes, which can be detected by households themselves. This result corroborates past work and, alongside the time savings described above, helps explain why households value clean water and are willing to spend or forego cash to acquire it, even though alternatives are available.

¹⁹Though we did not collect data on the identity of the household member who collects the water, it is extremely common in developing countries for this task to be borne overwhelmingly by women, so reductions in time spent collecting water may have gendered benefits (UNICEF 2017).

²⁰Note that the probability of reporting treatment in a given week is lower than the share of households that ever report using a given treatment technology. In the case of chlorine, for instance, while 13 percent of control households ever report treating their water with chlorine, the share reporting chlorine usage in the average survey round is only 3.2 percent. In this setting, we expect households to be aware of the benefits of chlorine, as local health workers train households on the use of chlorine as a water treatment practice (Ministry of Health and Family Welfare, Government of India 2006, 2021).

²¹In the exchangeable entitlement and free ration arms, point estimates suggest there may be declines in straining, though these are not statistically different from zero. While straining may reduce cholera and worms-based illnesses, it is ineffective for viruses, bacteria, and small protozoa such as giardia (World Health Organization 2008).

TABLE 2—INTENT-TO-TREAT EFFECTS OF WATER OFFERS ON WATER COLLECTION AND PURIFICATION

	Collection time (1)	Chlorinates (2)	Strains (3)	Boils (4)	Fuel time (5)	Fuel cost (6)
Prices (discounts)	−4.77 (1.81)	0.00 (0.01)	0.02 (0.03)	0.00 (0.02)	−1.99 (10.95)	−5.54 (10.55)
Exchangeable entitlement	−8.95 (2.12)	−0.01 (0.01)	−0.06 (0.04)	−0.01 (0.04)	−4.31 (5.81)	−6.07 (18.88)
Free ration	−12.55 (2.54)	−0.01 (0.01)	−0.04 (0.02)	−0.01 (0.02)	−18.32 (14.36)	−9.95 (12.63)
Observations	1,523	1,535	1,535	1,535	335	335
Control means	32.339	0.032	0.195	0.096	60.267	31.614

Notes: This table presents intent-to-treat effects of water offers on water collection and purification, estimated using equation (3). Column 1 is water collection time in minutes per day. Columns 2, 3, and 4 are binary indicators for treating water with chlorine, straining, or boiling, respectively. Column 5 is the amount of time spent collecting fuel for boiling in minutes per day, and column 6 is amount of money spent on fuel for boiling in rupees, both for only the households that report boiling water. We restrict the sample to village months where Spring Health delivered water. All regressions include village and month-of-sample fixed effects. Standard errors are clustered by village.

This is because it is perceived benefits that influence household demand.²² We use two-stage least squares to estimate the LATE of drinking clean water from Spring Health (not necessarily exclusively) on self-reported illness, health expenditures, and missed work due to sickness.

We estimate LATEs for each experimental arm, with first stages of

$$1[\text{Drinks clean water}]_{it} \times \text{Any discount}_i = \eta \cdot \text{Any discount}_i + \gamma_v + \delta_t + \varepsilon_{it}$$

$$1[\text{Drinks clean water}]_{it} \times \text{Any exchange}_i = \eta \cdot \text{Any exchange}_i + \gamma_v + \delta_t + \varepsilon_{it}$$

$$1[\text{Drinks clean water}]_{it} \times \text{Free ration}_i = \eta_1 \cdot \text{Free ration}_i + \gamma_v + \delta_t + \varepsilon_{it},$$

and a second stage of

$$\begin{aligned}
 (4) \quad Y_{it} = & \theta_1 \cdot \overline{1[\text{Drinks clean water}]_{it} \times \text{Any discount}_i} \\
 & + \theta_2 \cdot \overline{1[\text{Drinks clean water}]_{it} \times \text{Any exchange}_i} \\
 & + \theta_3 \cdot \overline{1[\text{Drinks clean water}]_{it} \times \text{Free ration}_i} \\
 & + \gamma_v + \delta_t + \varepsilon_{it},
 \end{aligned}$$

where $1[\text{Drinks clean water}]_{it}$ is an indicator for whether the household reports drinking any clean water, all other terms are defined as in equation (3), and all sample restrictions are identical. We use an indicator for any clean water consumption

²² We note, however, that clinical outcomes are what matter for policy. We therefore do not use these self-reported health impacts in the cost-effectiveness analysis below, but nevertheless view them as useful in understanding household choices.

as the endogenous variable rather than quantity because our primary object of interest is the effect of clean water access on health, rather than a dose-response function. Moreover, conditional on orders, water quantity is fairly homogeneous across households. Table 3 reports the results. The term “clean water” in these regression specifications (as in the remainder of the paper) refers only to the treated water of our intervention. This is because we are interested here in investigating whether this product, for which we observe substantial WTP and WTA, improves self-reported health.

We estimate that drinking clean water substantially reduces self-reported illness: The point estimates in the table correspond to reductions equivalent to 62 percent of control (0.3 fewer sick individuals reported per week on average, false discovery rate (FDR) adjusted $p = 0.03$), 36 percent (0.18 fewer sick people reported, FDR-adjusted $p = 0.09$), and 23 percent (not different from zero) in the discount, exchangeable entitlement, and free ration arms, respectively. The 95 percent confidence intervals include effects ranging from 6 to 58 percentage points (prices), 0 to 36 percentage points (exchange), and -6 to 30 percentage points (free ration). Drinking clean water meaningfully reduces weekly health expenditure, by 37 percent (Rs 91 per week on average, FDR-adjusted $p < 0.01$), 41 percent (Rs 97, FDR-adjusted $p = 0.11$), and 27 percent (Rs 64, not different from 0) across the discount, exchangeable entitlement, and free ration arms. We also find large negative IV effects of drinking clean water on missing work in the price and exchange arms: 12 percentage points (59 percent of control, FDR-adjusted $p = 0.02$), 13 percentage points (66 percent, FDR-adjusted $p = 0.07$), and 8 percentage points (40 percent, FDR-adjusted $p = 0.24$) in the discount, exchangeable entitlement, and free ration arms, respectively.

Again, we document evidence that clean water improves self-reported household health. Supplemental Appendix Table D.D3 also presents intent-to-treat effects of water offers on self-reported health, and includes a breakdown of illness into various symptoms, all of which are linked to drinking contaminated water (World Health Organization 2017). These also indicate improvements in self-reported health, though as expected, the LATEs are larger than the intent-to-treat estimates.

The impacts of clean water on self-reported sickness are largest and most precise in the price arm, while our estimated LATEs are smaller and less precisely estimated in the exchangeable entitlement arm and the free ration arm (though we are unable to reject equality between our IV effects across arms). To the extent there are differences in these impacts, they are likely driven by (unobservable) selection.²³ While nearly all households order clean water in the exchangeable entitlement and free ration arms, fewer households order water in the price arm, and this varies by discount level. Column 4 of Supplemental Appendix Table D.2 shows how the probability of households self reporting that they drank treated bottled water in the last week changes with treatment offers and discount levels. More compliers come from the lower price groups, but high-price groups also screen out take-up. The LATE compliers in the price arm, that must spend to order water, are likely to be the households that are

²³ Lee, Miguel, and Wolfram (2020a) document a similar phenomenon in the context of rural electrification: Households that purchase electricity connections at higher prices have larger benefits.

TABLE 3—LATE OF CLEAN WATER ON SELF-REPORTED HEALTH

	Sickness (1)	Health expenses (2)	Missed work (3)
Drinks SH water (prices)	−0.32 (0.13) [0.03]	−90.95 (24.86) [< 0.01]	−0.12 (0.04) [0.02]
Drinks SH water (exchange)	−0.18 (0.09) [0.09]	−97.07 (55.83) [0.11]	−0.13 (0.05) [0.07]
Drinks SH water (free ration)	−0.12 (0.09) [0.28]	−63.76 (56.03) [0.28]	−0.08 (0.04) [0.24]
Observations	5,271	5,219	5,271
Control means	0.516	239.51	0.2
Price = exchange	0.26	0.91	0.79
Price = ration	0.23	0.67	0.52
Exchange = ration	0.70	0.68	0.49

Notes: This table reports instrumental variable estimates of the effect of a household reporting drinking clean water on health outcomes, estimated using equation (4). In column 1, the outcome is the number of household members being sick in the past week. In column 2, the outcome is household spending on health in the past week in Indian rupees. In column 3, the outcome is an indicator for the number of household members missing work due to illness in the past week. We restrict the sample to village months where Spring Health delivered water, and drop the top 1 percent of health expenses to remove large outliers. The regression includes village and month-of-sample fixed effects. Standard errors are clustered by village. FDR-adjusted *p*-values in brackets. Price = exchange, Price = ration, and Exchange = ration are *p*-values on tests for equality between the price and exchange arm, price and ration arm, and exchange and ration arm, respectively.

at highest risk of falling ill—precisely the prediction from the selection model we describe in Section IVA.²⁴

Improvements in self-reported health outcomes (alongside self-reported time savings) help to rationalize households' relatively high valuation of home-delivered clean water. These results align with other papers that study self-reported health outcomes in the context of clean water interventions (Kremer et al. 2011b; Dupas et al. 2023). That said, self-reported health improvements are not the same as clinical benefits and, therefore, the cost-effectiveness calculations in Section V are not based on these results.

E. Treatment Effect Heterogeneity

In this section, we explore heterogeneity in our treatment effects in valuation and health.

Heterogeneous Valuation.—Due to extremely high take-up in the exchangeable entitlements arm, there is limited scope for heterogeneity in WTA. However, as take-up varies by offers in the prices arm, we can estimate heterogeneity in WTP.

²⁴It is also possible that households that face monetary costs for clean water are more likely to take care to keep it clean, or to engage in complementary sanitation behaviors, though we do not have direct evidence of this.

To do so, we split our main sample based on the few covariates that are determined before the experiment, reestimate our demand curves as described previously, and calculate the area under these curves to obtain population average WTP for each subsample. Specifically, we measure heterogeneity according to whether the household has (i) a child below five, (ii) a household head with more than primary education, (iii) above median income, or (iv) a piped water connection. Figure 5 shows the WTP estimates for each of these subgroups. We cannot reject equality in WTP for any subgroup, so we treat this heterogeneity as suggestive.

Though the differences in WTP are imprecisely estimated, the point estimates imply that WTP is meaningfully higher among households with above-median monthly income and for households with a piped water connection. It is unsurprising that households with higher incomes have higher WTP, and indeed, prior studies have shown that richer households are willing to pay higher amounts for clean air (Ito and Zhang 2020) and electricity (Lee, Miguel, and Wolfram 2020a). This may reflect the notion that WTP and ability to pay are in better alignment in high-income households (e.g., owing to the presence of liquidity constraints), or simply that clean water is a normal good, and poorer households would prefer to reserve their limited income for even more basic needs (Greenstone and Jack 2015).

We also see suggestive evidence that households with piped water have a higher WTP than households without, which is perhaps on first glance surprising. However, two simple explanations can rationalize this. First, in order to receive piped water, households in our sample must typically expend both effort and funds.²⁵ Thus, this dimension of heterogeneity could reflect the same income dimension as discussed above. Second, households that are willing to pay for a tap are likely to value the benefits of tap water—many of which are shared by Spring Health's model—highly, explaining their large WTP.

There is little evidence that the presence of a small child or education levels matters for household WTP. This might suggest that households value benefits for older children or adults as well, such as fewer days of missed work.

These results provide suggestive evidence that a piped water connection is not viewed by many households as a satisfactory substitute for clean drinking water. The evidence on water quality presented in Supplemental Appendix Figure A.2 suggests that piped water is not automatically a solution to the lack of clean drinking water from the standpoint of a technocratic planner. The evidence from WTP measures presented here suggests it is not seen as one by households either. It is important to note here that our results do not imply that piped water in itself is not a worthwhile investment, nor do they preclude high WTP or WTA for piped water (see, e.g., Devoto et al. 2012). There are many benefits to piped water even if it is not safe to drink, including convenience and access to a large volume of water for many end uses. Our results do, however, demonstrate that piped water alone may not meet the clean water needs of many consumers.

²⁵This normally consists of a subsidized connection fee and sometimes a subsidized monthly tariff. In the case of rural connections this may be paid to a village-level managing institution, in the case of municipal connections to a water utility.

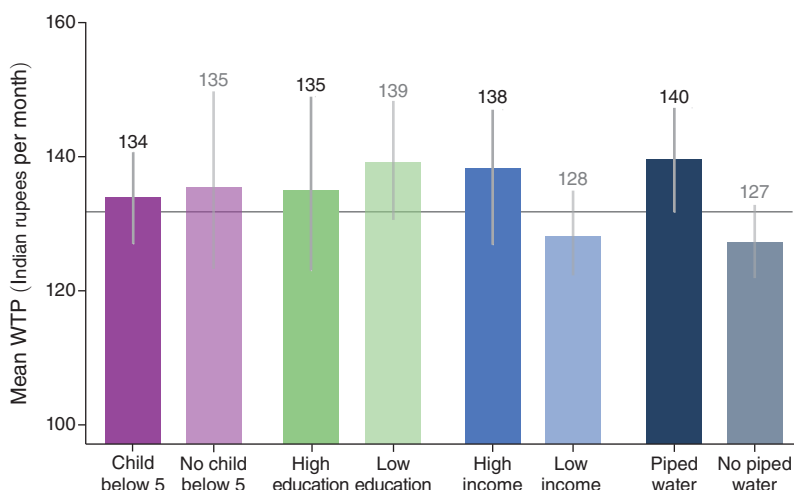


FIGURE 5. HETEROGENEITY IN WTP FOR CLEAN DRINKING WATER

Notes: Each bar shows the average WTP for the subset of survey households that report a particular value of four key covariates: the presence (or absence) of children below five (purple), whether the household head has (or has not) completed primary education (green); whether the household has above- (or below-) median income (light blue); and whether the household ever reports using piped water or not (navy blue). To compute WTP, we estimate subsample-specific demand curves, as in Figure 3, and calculate the area under the stepwise demand function, linearly interpolating demand between price points. Error bars show 95 percent confidence intervals of the mean WTP, estimated using a village-wise block bootstrap. The horizontal line shows the population-average WTP, calculated using the demand curve in Figure 3.

Heterogeneous Health Effects.—In Supplemental Appendix Table D.4, we present results examining heterogeneity in self-reported health outcomes. We do not see statistically significant differences in treatment effects by above versus below median education, piped water access, or above- versus below-median education. We do see statistically significant differences in treatment effects on all three self-reported health outcomes between households with and without children under five in the price arm. The relevant point estimates for the other treatment arms go in the same direction but are less precisely estimated. We take this as supportive evidence that our self-reported health effects are driven by clean water access, as children are likely the household members who are most susceptible to diseases such as diarrhoea, and the presence of ill children can easily lead parents to miss work.

V. Discussion

Drinking Water Policy and Household Valuation.—A key contribution of this experiment is we are able to estimate a lower bound of a WTA measure of the value of clean water via revealed preference. Our lower bound estimate of Rs 420 per month for clean water access is substantial, and suggests that households would be willing to forgo at least Rs 420 per month (or Rs 5,040 annually) for free access to clean water sufficient to cover all their drinking needs. This amount of money would more than cover the variable costs of clean water provision. Interestingly, Odisha's flagship cash transfer for rural farmers, Krushak Assistance for Livelihood and Income Augmentation, provided approximately Rs 1,700 per household per

month on launch (Dhillon 2019). It may therefore be possible for the government to repurpose some cash transfer funds to clean water provision, or even provide opt-in choices to households. Because Odisha is far from alone in using large-scale cash transfers, such a policy may be worth testing more widely.²⁶

Scaling Clean Water Delivery.—In this experiment, we compute the partial equilibrium impacts of clean water access in 120 villages in which Spring Health had been operating prior to the study. However, solving the global clean water access problem will require interventions that can be delivered at scale. While our experiment does not speak to the logistics of scale directly, and though these villages may be different from locations where no such firm already exists, features of our intervention suggest that this may be feasible. First, our partner organization, Spring Health, is a private company, which has been delivering clean water in Odisha for more than ten years. Second, we use cost data from Spring Health to compute the net present value of profits under different price levels (see Supplemental Appendix G for full details). We plot the results in Figure 6.

Figure 6 reveals that home delivery of clean water can be privately profitable. Moreover, under the status quo price of Rs 1.4 per liter, Spring Health is only just below the profitability cutoff we estimate. Spring Health augments its water sales income with other revenue streams, including the sale of carbon credits generated by treating water using solar electricity.

However, offering the low prices required for *universal* clean water access would not be privately profitable. Doing so would therefore require subsidies, but our results suggest that governments would not need to cover the fixed costs of water treatment plants. Even were they to do so, perhaps in the interests of scaling up rapidly, the costs are substantially lower than building an extensive piped water network. As a result, we expect that home delivery of clean water is likely to be feasible to scale.

Indeed, the private provision of clean water is proliferating in low- and middle-income countries (Cohen and Ray 2018), with decentralized water kiosk and home delivery models like that used by Spring Health becoming increasingly common (Advani et al. 2011; Brown et al. 2011; Daly et al. 2021). The growth of nontap alternatives reflects some attractive characteristics of this source of clean water. Decentralized treatment and home delivery of drinking water has much lower capital and maintenance costs than piped water, such that it can even be profitably supplied by private providers like our implementation partner. This allows the possibility of rapidly reaching geographies or households that do not yet have safe piped water.

Cost-Effectiveness.—Our data suggest that providing clean water at high prices can be privately profitable. However, our estimated demand curve predicts that only a small fraction of households will buy water at these prices. These results therefore explain why the private market has not fully solved the universal clean drinking water access problem. However, state spending on most preventive health interventions

²⁶Cash transfer programs exist in more than 120 low- and middle-income countries (Banerjee et al. 2024), covering more than 1 billion people around the world (Niehaus and Suri forthcoming). India is no exception, with flagship central schemes such as Pradhan Mantri Kisan Samman Nidhi for farmers and Pradhan Mantri Matru Vandana Yojana for expectant mothers (Weaver et al. 2024).

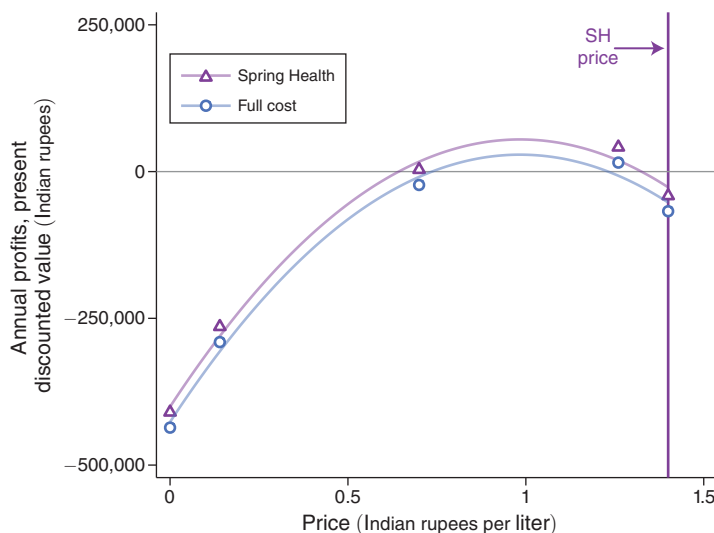


FIGURE 6. PROFITABILITY OF CLEAN WATER AT DIFFERENT PRICES

Notes: This figure plots the present discounted value of annual profits from home delivery of clean water. We assume a clean water plant has a ten year lifespan, a discount rate of 5 percent, an average village size of 450 households, and average consumption among water buyers of 237 L per month (per the control group in the price regime), and fit the zero price point using take-up in the free ration arm. We use our estimated extensive margin demand curve, derived from the price arm, to calculate the share of households that order at each price level, which affects total revenue and total variable costs. In light blue circles (“full cost”), we plot profits at each price level calculated as total revenue less one-time capital costs, monthly fixed costs, and total variable costs. In purple triangles (“Spring Health”), we subtract off capital costs, as Spring Health receives donor funding to cover these costs. Curves are quadratic fits. Finally, we plot the Spring Health status quo price (Rs 1.4 per liter) as a purple vertical line.

is not based on WTP alone. Rather, the cost-effectiveness of such interventions is typically tested by quantifying expected gains in DALYs, and comparing the cost per DALY to standard benchmarks (e.g., $1 \times$ per capita GDP from World Health Organization 2020 or \$500 for low- and middle-income countries from Jamison et al. 2018). This approach reflects a social planner who places a high value on universal safe drinking water, due to positive externalities such as reduced waterborne disease transmission. By using GDP rather than wages, the WHO benchmark also seeks to capture societal multiplier effects.

We therefore calculate the cost per DALY saved and the net benefits of clean water delivery to measure cost-effectiveness. The key inputs into this calculation are (i) the take-up of clean water when it is provided for free, which we estimate using our free ration arm; (ii) the impact of clean water on infant mortality, which we draw from the Kremer et al. (2023) meta-analysis; and (iii) the cost of providing clean water, which we compute using our data. We calculate cost-effectiveness for Spring Health water, under varying assumptions (subsidies for all households versus households with children; government covering only variable or variable and fixed costs; and water availability per our experiment or 100 percent of the time) and for two chlorine-based approaches (chlorine dispensers and maternal-child-health center coupon distribution), where we take the key parameters from Kremer et al.

(2023), but adjust to the Odisha context. We discuss these calculations in full in Supplemental Appendix G.

Figure 7 presents the results. The top panel plots the cost per DALY under each scenario, compared to standard benchmarks. We estimate a cost per DALY of Spring Health water ranging from \$71 to \$226. In this setting, dispensers cost \$106 per DALY, and chlorine coupons cost just \$33. All five measures are far below WHO- and World Bank-proposed cost-effectiveness thresholds. The bottom panel plots net benefits per child, which are high in all scenarios: \$894 in welfare gains per child from dispensers, \$673 from chlorine coupons, and \$1,821 to \$2,256 from Spring Health water. The net benefits from Spring Health water are meaningfully higher than from chlorine in this setting due to large differences in take-up: 89 percent–90 percent for Spring Health versus 36 percent for dispensers and 26 percent for coupons, even though the cost of provision per child is meaningfully lower for chlorine than Spring Health water. All of the water sources we study would remain cost effective according to the more stringent benchmark even if costs were doubled (or benefits halved), and would continue to satisfy the less stringent benchmark up until costs increase eightfold, suggesting that this policy result is likely insensitive to study-specific modeling choices or particularities of our sample. Broadly, these results demonstrate that both chlorine and home delivery of clean water pass a cost-benefit test with flying colors, and highlight the value of increasing take-up of clean water.

Mechanisms for Allocating Water.—Finally, our results shed light on whether policymakers should use rations or price mechanisms to allocate clean water, in the tradition of Weitzman (1977). While standard economic theory tells us that prices optimally allocate resources, rations and in-kind transfers are widely used: More than 90 percent of low-income countries' social safety nets include in-kind transfers (Gentilini, Honorati, and Yemtsov 2014). These zero-price mechanisms are used to distribute a series of subsistence goods, including food (Cunha, de Giorgi, and Jayachandran 2019; Gadenne 2020; Gadenne et al. 2024) and electricity (Jack and Smith 2020). While determining which approach is preferred is a complicated public finance question (Gadenne and Singhal forthcoming), the classic argument against rationing is that (i) rations fail to allocate scarce resources to those with the highest demand, and (ii) rations lead to waste.

In our setting, preferences for clean drinking water appear to be stable and relatively homogenous, and we find little evidence of waste under free distribution, in the sense that households order less than the ration limit. Under these conditions, free distribution may be a reasonably efficient way of allocating clean water to the poor. This approach can be logistically easier than subsidized pricing and our experiment suggests households value water highly enough to justify such a policy. Furthermore, over and above household valuation, there are arguably positive externalities to ensuring universal access to clean drinking water (e.g., reduced transmission of waterborne diseases).

That said, from Table 3, our LATEs on health appear to be strongest in the discount group. This suggests that higher-WTP households may enjoy larger health benefits from clean water (as also shown for rural electrification in Lee, Miguel, and Wolfram 2020b). Replicating and investigating this result in future studies would be valuable, especially because other important preventive health investments have not shown

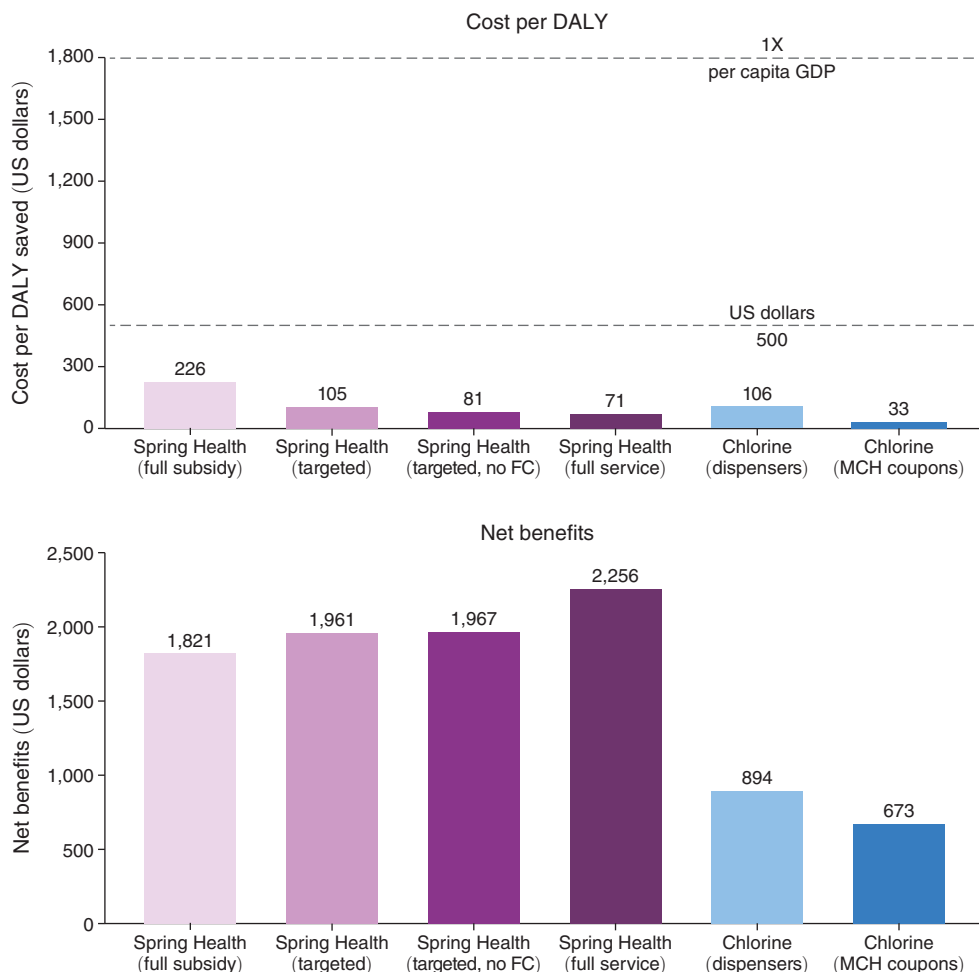


FIGURE 7. COST-EFFECTIVENESS OF CLEAN WATER

Notes: This figure presents cost-effectiveness measures of clean water under six scenarios: Spring Health (full subsidy), simulating entering a village as-yet-unserved by Spring Health, providing 100 percent clean water subsidies to all households, and covering fixed costs; Spring Health (targeted), simulating entering a village as yet-unserved by Spring Health, providing 100 percent clean water subsidies only to households with children under five, and covering fixed costs; Spring Health (targeted, no FC), simulating providing 100 percent clean water subsidies to households with children under five in a village already served by Spring Health and, therefore, excluding fixed costs; Spring Health (full service), same as (targeted, no FC) but simulating Spring Health water being available in 100 percent of months rather than 87.7 percent; chlorine (dispensers), simulating providing point-of-collection chlorine treatment; and chlorine (MCH coupons), simulating providing coupons for point-of-use chlorine treatment via maternal and child health services. All five scenarios are set in the context of rural Odisha, using local GDP per capita, infant mortality, and children per household. The top panel plots costs per DALY saved, with a WHO benchmark ($1 \times$ GDP per capita) and a World Bank LMIC benchmark (\$500) represented by gray dashed lines. The bottom panel plots net benefits per child, calculated as DALYs saved (evaluated at $1 \times$ per capita GDP) less the cost of provision. See Supplemental Appendix G for more details.

Sources: Kremer et al. (2011b); Dupas et al. (2016)

higher benefits when sold rather than given away (Cohen and Dupas 2010). One reason why water may be different is that maximizing the health benefits of drinking water likely requires complementary investments from households, including proper sanitation behavior and keeping utensils and containers clean.

VI. Conclusion

In this paper, we conduct a randomized trial to study a novel approach to addressing the global clean water access problem: home delivery of safe, pleasant-tasting drinking water. We use this experiment to provide two valuation measures. We produce what are, to the best of our knowledge, the first direct experimental revealed-preference measures of both households' WTP and WTA for clean water.

Both our WTP and our WTA measures demonstrate that households value clean drinking water highly. Our estimated WTP is several times higher than previous work on water in the literature, which has either estimated this quantity indirectly or instead observed demand for point-of-use purification as opposed to clean water. We show that a lower-bound on the WTA is higher still, consistent with classic theoretical predictions for valuation in the absence of substitutes. Our results highlight both the importance of directly measuring the good in question when valuing environmental quality and the importance of measuring both WTP and WTA in order to interpret consumer demand in environmental and development settings.

In addition to our findings on valuation, our intervention generated near universal and sustained take-up at low prices. Although we are unaware of a similar evaluation in the literature, our intervention is informed by an emerging private market for clean water delivery targeted at the poor. We therefore suggest that decentralized treatment and home delivery of clean water should be given serious consideration as a solution to arguably one of the most important health risks of our time—lack of access to clean drinking water.

This approach to increasing clean water access appears to be both sustainable and scaleable. Our cost-effectiveness analysis suggests that free provision of clean water would be strongly welfare improving. Moreover, cost data from our implementation partner suggests that water sales using decentralized treatment and home delivery can even be privately profitable at high prices. As a result, governments could likely scale clean water access substantially simply by providing rebate vouchers to private providers in order to reduce consumer prices to zero.

There are important benefits to the two approaches most commonly studied to date. Chlorine treatment at home is still the cheapest way to remove coliform from water and tablets and solutions are easy to distribute even in very remote areas. Piped water is very convenient, and if it can be kept clean, it has the potential to ensure all water used by the household is safe, limiting spillover contamination. However, it is clear that neither option is a universal solution in the short or medium term. Consequently, it would be very valuable to build a greater body of evidence on directly and conveniently providing drinking water at home, cleaned using decentralized treatment.

REFERENCES

- Advani, Rajesh K., Malva Rosa Baskovich, Sabrina Birner, Aurelien Antoine Claude Boyer, Will Davies, Malak Draz, Leila H. Elvas, et al. 2011. *Bringing Water to Where it is Needed Most: Innovative Private Sector Participation in Water and Sanitation*. World Bank Group.
- Ahuja, Amrita, Michael Kremer, and Alix Peterson Zwane. 2010. "Providing Safe Water: Evidence from Randomized Evaluations." *Annual Review of Resource Economics* 2 (1): 237–56.
- Alsan, Marcella, and Claudia Goldin. 2019. "Watersheds in Child Mortality: The Role of Effective Water and Sewerage Infrastructure, 1880–1920." *Journal of Political Economy* 127 (2): 586–638.

- Amiran, Edoh Y., and Daniel A. Hagen. 2003. "Willingness to Pay and Willingness to Accept: How Much can They Differ? Comment." *American Economic Review* 93 (1): 458–63.
- Ashraf, Nava, James Berry, and Jesse M. Shapiro. 2010. "Can Higher Prices Stimulate Product Use? Evidence from a Field Experiment in Zambia." *American Economic Review* 100 (5): 2383–2413.
- Atmashakti Trust. 2023. *Safe Drinking Water: An Alarming Situation*. <https://www.atmashaktitrust.org/wp-content/uploads/2024/10/Safe-Drinking-Water-Report.pdf> (accessed February 9, 2026)
- Banerjee, Abhijit, Rema Hanna, Benjamin A. Olken, and Diana Sverdlin Lisker. 2024. "Social Protection in the Developing World." *Journal of Economic Literature* 62 (4): 1349–1421.
- Baylis, Patrick, Michael Greenstone, Kenneth Lee, and Harshil Sahai. 2024. "Is the Demand for Clean Air Too Low? Experimental Evidence from Delhi." Unpublished.
- Becker, Gordon M., Morris H. DeGroot, and Jacob Marschak. 1964. "Measuring Utility by a Single-Response Sequential Method." *Behavioral Science* 9 (3): 262–32.
- Berkouwer, Susanna B., and Joshua T. Dean. 2022. "Credit, Attention, and Externalities in the Adoption of Energy Efficient Technologies by Low-Income Households." *American Economic Review* 112 (10): 3291–3330.
- Berry, James, Greg Fischer, and Raymond Guiteras. 2020. "Eliciting and Utilizing Willingness to Pay: Evidence from Field Trials in Northern Ghana." *Journal of Political Economy* 128 (4): 1436–73.
- Biswas, Ramakanta. 2022. "Majority of Rivers' Water Toxic in Odisha, Gangua Nala Most Polluted." Odisha TV. October 7. <https://odishatv.in/news/miscellaneous/majority-of-rivers-water-toxic-in-odisha-gangua-nala-most-polluted-187293>.
- Brown, Joe, Thomas Clasen, Tom Outlaw, Mark D. Sobsey, and Jianyong Wu. 2011. *Safe Water for All: Harnessing the Private Sector to Reach the Underserved*. World Bank Group.
- Burlig, Fiona, Amir Jina and Anant Sudarshan. 2023. "Liquidity constraints: Allocating clean water to the rural poor." AEA RCT Registry. <https://doi.org/10.1257/rct.10545-2.0>.
- Burlig, Fiona, Amir Jina, and Anant Sudarshan. 2026. *Data and Code for: "The Value of Clean Water: Experimental Evidence from Rural India"*. American Economic Association; distributed by Inter-university Consortium for Political and Social Research <https://doi.org/10.3886/E238281V1>.
- Centers for Disease Control and Prevention. 2021. *Eat and Drink Safely*. Centers for Disease Control and Prevention. [https://wwwnc.cdc.gov/travel/destinations/traveler/none/\[COUNTRY\]](https://wwwnc.cdc.gov/travel/destinations/traveler/none/[COUNTRY]). (Scraped via the Wayback Machine by the authors).
- Chen, Jiafeng, and Jonathan Roth. 2024. "Logs with Zeros? Some Problems and Solutions." *Quarterly Journal of Economics* 139 (2): 891–936.
- Cilliers, Jacobus, Nour Elashmawy, and David McKenzie. 2024. "Using Post-Double Selection Lasso in Field Experiments." Unpublished.
- Clasen, Thomas F., Kelly T. Alexander, David Sinclair, Sophie Boisson, Rachel Peletz, Howard H. Chang, Fiona Majorin, and Sandy Cairncross. 2015. *Interventions to Improve Water Quality for Preventing Diarrhoea*. *Cochrane Database of Systematic Reviews* 10: 10.1002/14651858.CD004794.pub3.
- Cohen, Alasdair, and Isha Ray. 2018. "The Global Risks of Increasing Reliance on Bottled Water." *Nature Sustainability* 1: 327–29.
- Cohen, Jessica, and Pascaline Dupas. 2010. "Free Distribution or Costsharing? Evidence from a Randomized Malaria Prevention Experiment." *Quarterly Journal of Economics* 125 (1): 1–45.
- Conroy, Ronàn, Michael Elmore-Meegan, Tina Joyce, Kevin G. McGuigan, and Joseph Barnes. 1996. "Solar Disinfection of Drinking Water and Diarrhoea in Maasai Children: A Controlled Field Trial." *Lancet* 348 (9043): 1695–97.
- Crider, Yoshika, Sonia Sultana, Leanne Unicomb, Jennifer Davis, Stephen P. Luby, and Amy J. Pickering. 2018. "Can You Taste It? Taste Detection and Acceptability Thresholds for Chlorine Residual in Drinking Water in Dhaka, Bangladesh." *Science of the Total Environment* 613: 840–46.
- Cunha, Jesse M., Giacomo de Giorgi, and Seema Jayachandran. 2019. "The Price Effects of Cash Versus In-Kind Transfers." *Review of Economic Studies* 86 (1): 240–81.
- Daly, Sean W., Jeremy Lowe, Gracie M. Hornsby, and Angela R. Harris. 2021. "Multiple Water Source use in Low- and Middle-Income Countries: A Systematic Review." *Journal of Water and Health* 19 (3): 370–92.
- Development Data Lab. 2025. *The Socioeconomic High-resolution Rural—Urban Geographic Platform for India (SHRUG)*. <https://www.devdatalab.org/shrug>.
- Devoto, Florencia, Esther Duflo, Pascaline Dupas, William ParientÅLe, and Vincent Pons. 2012. "Happiness on Tap: Piped Water Adoption in Urban Morocco." *American Economic Journal: Economic Policy* 4 (4): 68–99.
- Dhillon, Dilsher. 2019. "Here's what Modi government can learn from Odisha's direct cash transfer scheme." *Business Insider India*. https://www.businessinsider.in/direct-cash-transfer-scheme-in-odisha-is-a-lesson-for-modi-ahead-of-interim-budget/amp_articleshow/67719605.cms (accessed August 16, 2024).

- dos Santos, Carolina Tojal R., and Bruna Morais Guidetti. 2024. "Expansion of Piped Water and Sewer Networks: The Effects of Regulation." Unpublished.
- Dupas, Pascaline, and Edward Miguel. 2017. "Impacts and Determinants of Health Levels in Low-Income Countries." In *Handbook of Economic Field Experiments*, edited by Abhijit Banerjee and Esther Dufo, 3–93. North-Holland.
- Dupas, Pascaline, Basimenye Nhlema, Zachary Wagner, Aaron Wolf, and Emily Wroe. 2023. "Expanding Access to Clean Water for the Rural Poor: Experimental Evidence from Malawi." *American Economic Journal: Economic Policy* 15 (1): 272–305.
- Dupas, Pascaline, Vivian Hoffmann, Michael Kremer, and Alix Peterson-Zwane. 2016. "Targeting Health Subsidies through a Nonprice Mechanism: A Randomized Controlled Trial in Kenya." *Science* 353 (6302): 889–95.
- EFSA Panel on Dietetic Products, Nutrition, and Allergies. 2010. "Scientific Opinion on Dietary Reference Values for Water." *EFSA Journal* 8 (3): 1459.
- Ericson, Keith M. Marzilli, and Andreas Fuster. 2014. "The Endowment Effect." *Annual Review of Economics* 6: 555–79.
- Express News Service. 2024a. "BJP slams Odisha government over drinking water crisis." *New Indian Express*, April 30. <https://www.newindianexpress.com/states/odisha/2024/Apr/30/bjp-slams-odisha-government-over-drinking-water-crisis>.
- Express News Service. 2024b. "Thirsty villagers in Odisha allege neglect, stage dharna demanding drinking water." *New Indian Express*, April 2. <https://www.newindianexpress.com/states/odisha/2024/Apr/02/thirsty-villagers-in-odisha-allege-neglect-stage-dharna-demanding-drinking-water>.
- Gadenne, Lucie. 2020. "Can Rationing Increase Welfare? Theory and an Application to India's Ration Shop System." *American Economic Journal: Applied Economics* 12 (4): 144–77.
- Gadenne, Lucie, and Monica Singhal. Forthcoming. "The Form of Transfers: Cash, In-Kind, or Vouchers?" In *Handbook on Social Protection*, edited by Rema Hanna and Ben Olken. MIT Press.
- Gadenne, Lucie, Samuel Norris, Monica Singhal, and Sandip Sukhtankar. 2024. "In-Kind Transfers as Insurance." *American Economic Review* 114 (9): 2861–97.
- Galiani, Sebastian, Paul Gertler, and Ernesto Schargrofsky. 2005. "Water for Life: The Impact of the Privatization of Water Services on Child Mortality." *Journal of Political Economy* 113 (1): 83–120.
- Gamper-Rabindran, Shanti, Shakeeb Khan, and Christopher Timmins. 2010. "The Impact of Piped Water Provision on Infant Mortality in Brazil: A Quantile Panel Data Approach." *Journal of Development Economics* 92 (2): 188–200.
- Gentilini, Ugo, Maddalena Honorati, and Ruslan Yemtsov. 2014. *The State of Social Safety Nets 2014*. World Bank Group.
- Government of India. 2011. *Primary Census Abstract, India & States/UTs -State and district level, 2011*. Government of India.
- Government of India. 2023. *State-Wise Data on Per Capita Income*. Government of India.
- Greenstone, Michael, and B. Kelsey Jack. 2015. "Envirodevonomics: A Research Agenda for an Emerging Field." *Journal of Economic Literature* 53 (1): 5–42.
- Hanemann, W. Michael. 1991. "Willingness to Pay and Willingness to Accept: How Much Can They Differ?" *American Economic Review* 81 (3): 635–47.
- Hanemann, W. Michael. 2003. "Willingness to Pay and Willingness to Accept: How Much Can They Differ? Reply." *American Economic Review* 93 (1): 464.
- Haushofer, Johannes, Michael Kremer, Ricardo Maertens, and Brandon Joel Tan. 2021. "Water Treatment and Child Mortality: Evidence from Kenya." NBER Working Paper 29447.
- Horowitz, John K., and Kenneth E. McConnell. 2002. "A Review of WTA/WTP Studies." *Journal of Environmental Economics and Management* 44 (3): 426–47.
- Ito, Koichiro, and Shuang Zhang. 2020. "Willingness to Pay for Clean Air: Evidence from Air Purifier Markets in China." *Journal of Political Economy* 128 (5): 1627–72.
- Jack, Kelsey, and Grant Smith. 2020. "Charging Ahead: Prepaid Metering, Electricity Use, and Utility Revenue." *American Economic Journal: Applied Economics* 12 (2): 134–68.
- Jalan, Jyotsna, and E. Somanathan. 2008. "The Importance of being Informed: Experimental Evidence on Demand for Environmental Quality." *Journal of Development Economics* 87 (1): 14–28.
- Jal Jeevan Mission. 2022. *Functionality Assessment of Household Tap Connection under National Jal Jeevan Mission - 2022*. Government of Odisha.
- Jamison, Dean T., Hellen Gelband, Susan Horton, Prabhat Jha, Ramanan Laxminarayan, Charles N. Mock, and Rachel Nugent, ed. 2018. *Disease Control Priorities: Improving Health and Reducing Poverty (Third Edition)*. World Bank Group.
- Jeuland, Marc, Jennifer Orgill, Ameer Shaheed, Geoff Revell, and Joe Brown. 2016. "A Matter of Good Taste: Investigating Preferences for in-House Water Treatment in Peri-Urban Communities in Cambodia." *Environment and Development Economics* 21 (3): 291–317.

- Jeuland, Marc, Marcella McClatchey, Sumeet R. Patil, Subhrendu K. Pattanayak, Christine M. Poulos, and Jui-Chen Yang. 2021. "Do Decentralized Community Treatment Plants Provide Clean Water? Evidence from Rural Andhra Pradesh, India." *Land Economics* 97 (2): 345–71.
- Kremer, Michael, Edward Miguel, Sendhil Mullainathan, Clair Null, and Alix Peterson Zwane. 2011a. "Social Engineering: Evidence from a Suite of Take-Up Experiments in Kenya." Unpublished.
- Kremer, Michael, Gautam Rao, and Frank Schilbach. 2019. "Behavioral Development Economics." In *Handbook of Behavioral Economics - Foundations and Applications 2 Vol. 2*, edited by B. Douglas Bernheim, Stefano DellaVigna, and David Laibson, 345–458. North-Holland.
- Kremer, Michael, Jessica Leino, Edward Miguel, and Alix Peterson Zwane. 2011b. "Spring Cleaning: Rural Water Impacts, Valuation, and Property Rights Institutions." *Quarterly Journal of Economics* 126 (1): 145–205.
- Kremer, Michael, Stephen P. Luby, Ricardo Maertens, Brandon Tan, and Witold Wiecek. 2023. "Water Treatment and Child Mortality: A Meta-Analysis and Cost-Effectiveness Analysis." NBER Working Paper 30835.
- Lee, Kenneth, Edward Miguel, and Catherine Wolfram. 2020a. "Does Household Electrification Supercharge Economic Development?" *Journal of Economic Perspectives* 34 (1): 122–44.
- Lee, Kenneth, Edward Miguel, and Catherine Wolfram. 2020b. "Experimental Evidence on the Economics of Rural Electrification." *Journal of Political Economy* 128 (4): 1523–65.
- Luoto, Jill, Nusrat Najnin, Minhaj Mahmud, Jeff Albert, M. Sirajul Islam, Stephen Luby, Leanne Unicom, and David I. Levine. 2011. "What Point-Of-Use Water Treatment Products Do Consumers Use? Evidence From a Randomized Controlled Trial Among the Urban Poor in Bangladesh." *PLoS One* 6 (10): e26132.
- Ministry of Health and Family Welfare. 2022. *National Family Health Survey (NFHS-5), 2019–21, State fact sheet*. Odisha. Government of India.
- Ministry of Health and Family Welfare, Government of India. 2006. *Reading Material for ASHA – Book No. 1*. National Health Mission. Ministry of Health and Family Welfare, Government of India.
- Ministry of Health and Family Welfare, Government of India. 2021. *Induction Training Module for ASHA (English)*. National Health Systems Resource Centre. Ministry of Health and Family Welfare, Government of India.
- Ministry of JalShakti. 2022. *Contamination of Ground Water*. <https://pib.gov.in/PressReleaseIframePage.aspx?PRID=1809264> (accessed February 6, 2026)
- MOSPI. 2021. *Gendering Human Development: A Working Paper for Computing HDI, GDI, and GII for states of India*. National Statistics Office, Government of India.
- MOSPI. 2024. *Household Consumption Expenditure Survey: 2022-23 Fact Sheet*. National Sample Survey Office, Government of India.
- National Sample Survey Office. 2023. *NSS 78th round: Multiple Indicator Survey in India, 2020–2021*. National Sample Survey Report 589.
- Niehaus, Paul, and Tavneet Suri. Forthcoming. "Cash Transfers." In *Handbook on Social Protection*, edited by Rema Hanna and Ben Olken. MIT Press.
- Null, Clair, Christine P. Stewart, Amy J. Pickering, Holly N. Dentz, Benjamin F. Arnold, Charles D. Arnold, Jade Benjamin-Chung, et al. 2018. "Effects of Water Quality, Sanitation, Handwashing, and Nutritional Interventions on Diarrhoea and Child Growth in Rural Kenya: A Cluster-Randomised Trial." *Lancet Global Health* 6 (3): e316–29.
- Null, Clair, Michael Kremer, Edward Miguel, Jorge Garcia Hombrados, Robyn Meeks, and Alix Peterson Zwane. 2012. "Willingness-to-Pay for Cleaner Water in Less Developed Countries: Systematic Review of Experimental Evidence." *3ie Systematic Review* 006.
- Odagiri, Mitsunori, Alexander Schriewer, Miles E. Daniels, Stefan Wuertz, Woutrina A. Smith, Thomas Clasen, Wolf-Peter Schmidt, Yujie Jin, Belen Torondel, Pravas R. Misra, Pinaki Panigrahi, and Marion W. Jenkins. 2016. "Human Fecal and Pathogen Exposure Pathways in Rural Indian Villages and the Effect of Increased Latrine Coverage." *Water Research* 100: 232–44.
- Pattanayak, Subhrendu K., Jui-Chen Yang, Dale Whittington, and K.C. Bal Kumar. 2005. "Coping with Unreliable Public Water Supplies: Averting Expenditures by Households in Kathmandu, Nepal." *Water Resources Research* 41 (2).
- Plott, Charles R., and Kathryn Zeiler. 2005. "The Willingness to Pay Willingness to Accept Gap, The 'Endowment Effect,' Subject Misconceptions, and Experimental Procedures for Eliciting Valuations." *American Economic Review* 95 (3): 530–45.
- Puget, Sabine, Noëlle Beno, Claire Chabanet, Elisabeth Guichard, and Thierry Thomas-Danguin. 2010. "Tap Water Consumers Differ from Non-Consumers in Chlorine Flavor Acceptability but Not Sensitivity." *Water Research* 44 (3): 956–64.

- Reese, Heather.** 2017. "Effectiveness of a Combined Sanitation and household Level Piped Water Intervention on Infrastructure Coverage, Availability and Use, Environmental Fecal Contamination, and Child Health in Rural Odisha, India: A Matched Cohort Study." PhD diss. Emory University.
- Rukmini S.** 2024. *Access to Drinking Water*. <https://www.dataforindia.com/access-to-drinking-water> (accessed October 20, 2024).
- Senapati, Ashis.** 2021. "Odisha has 19 of 351 polluted rivers in country, says Union minister Gajendra Singh Shekhawat." *Times of India*, July 28. <https://timesofindia.indiatimes.com/city/bhubaneswar/odisha-has-19-of-351-polluted-rivers-in-country-says-union-minister-gajendra-singh-shekhawat/articleshow/84819565.cms>.
- Shogren, Jason F., and Dermot J. Hayes.** 1997. "Resolving Differences in Willingness to Pay and Willingness to Accept: Reply." *American Economic Review* 87 (1): 241–44.
- Shogren, Jason F., Seung Y. Shin, Dermot J. Hayes, and James B. Kliebensten.** 1994. "Resolving Differences in Willingness to Pay and Willingness to Accept." *American Economic Review* 84 (1): 255–70.
- Smith, Daniel W., Mahfuza Islam, Kirin E. Furst, Shobnom Mustaree, Yoshika S. Crider, Nazrin Akter, Syed Anjerul Islam, et al.** 2021. "Chlorine Taste Can Increase Simulated Exposure to Both Fecal Contamination and Disinfection Byproducts in Water Supplies." *Water Research* 207: 117806.
- Szabó, Andrea.** 2015. "The Value of Free Water: Analyzing South Africa's Free Basic Water Policy." *Econometrica* 83 (5): 1913–61.
- Tarozzi, Alessandro, Aprajit Mahajan, Brian Blackburn, Dan Kopf, Lakshmi Krishnan, and Joanne Yoong.** 2014. "Micro-loans, Insecticide-Treated Bednets, and Malaria: Evidence from a Randomized Controlled Trial in Orissa, India." *American Economic Review* 104 (7): 1909–41.
- Tuncel, Tuba, and James K. Hammitt.** 2014. "A New Meta-analysis on the WTP/WTB Disparity." *Journal of Environmental Economics and Management* 68 (1): 175–87.
- UNICEF.** 2017. *Progress on Drinking Water, Sanitation and Hygiene: 2017 Update and SDG Baselines*. World Health Organization and the United Nations Children's Fund.
- United Nations.** 2022. *The Sustainable Development Goals Report 2022*. United Nations.
- Weaver, Jeffrey, Sandip Sukhtankar, Paul Niehaus, and Karthik Muralidharan.** 2024. "Cash Transfers for Child Development: Experimental Evidence from India." NBER Working Paper 32093.
- Weitzman, Martin.** 1977. "Is the Price System or Rationing More Effective in Getting a Commodity to Those Who Need It Most?" *Bell Journal of Economics* 8 (2): 517–24.
- WHO, UNICEF.** 2024. *WHO/UNICEF Joint Monitoring Programme for Water Supply, Sanitation, and Hygiene*. World Health Organization.
- WHO, UNICEF, World Bank.** 2022. *State of the World's Drinking Water: An Urgent Call to Action to Accelerate Progress on Ensuring Safe Drinking Water for All*. World Health Organization.
- World Bank.** 2016. *High and Dry: Climate Change, Water, and the Economy*. World Bank.
- World Bank.** 2018. *Reducing Inequalities in Water Supply, Sanitation, and Hygiene in the Era of the Sustainable Development Goals: Synthesis Report of the WASH Poverty Diagnostic Initiative*. World Bank.
- World Bank.** 2024. *Funding A Water-Secure Future: An Assessment of Global Public Spending*. World Bank.
- World Health Organization.** 2008. *Guidelines for Drinking-Water Quality: Second Addendum to Third Edition*. World Health Organization.
- World Health Organization.** 2017. *Guidelines for Drinking-Water Quality, Fourth Edition, Incorporating the First Addendum*. World Health Organization.
- World Health Organization.** 2020. *WHO Methods and Data Sources for Global Burden of Disease Estimates, 2000–2019*. World Health Organization.